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Modul 7: Periodic quantities

Stand: 18. Februar 2026

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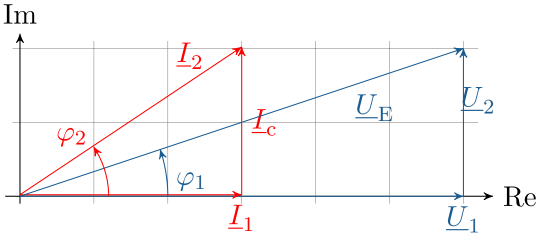
A sine wave has different voltage values over time. If the exact sine waves repeat themselves, it is a periodically recurring process. Module 7, "Periodic quantities", covers the following topics:

- Pointer diagram
- Complex alternating current calculation
- RMS value
- Apparent power, active power and reactive power
- Three-phase current
- Co-system, anti-system and null system



Learning objectives: Complex numbers

The Students can

- handle numbers in the complex plane.
- Plot phasor diagrams of complex numbers.
- calculate complex numbers.

Complex number plane

In the complex number plane, the imaginary unit j is introduced in order to perform calculations with complex numbers.

$$\mathrm{j} = \sqrt{-1}$$

Complex number plane

In the complex number plane, the imaginary unit j is introduced in order to perform calculations with complex numbers.

$$\mathrm{j} = \sqrt{-1}$$

Squaring the imaginary unit again gives -1.

$$\mathrm{j}^2 = -1$$

Representation of complex numbers

Cartesian coordinates with the real part and the product of the imaginary unit and the imaginary part:

$$\underline{Z} = Realpart + \mathrm{j} \cdot Imaginary\ part$$

Representation of complex numbers

Cartesian coordinates with the real part and the product of the imaginary unit and the imaginary part:

$$\underline{Z} = Realpart + \mathrm{j} \cdot Imaginary\ part$$

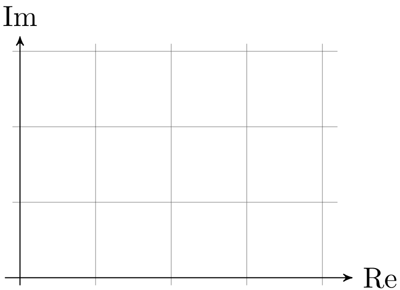
The real part is abbreviated with Re and the imaginary part with Im:

$$\underline{Z} = \mathrm{Re}(\underline{Z}) + \mathrm{j} \cdot \mathrm{Im}(\underline{Z})$$

Representation of complex numbers

Use of the complex layer:

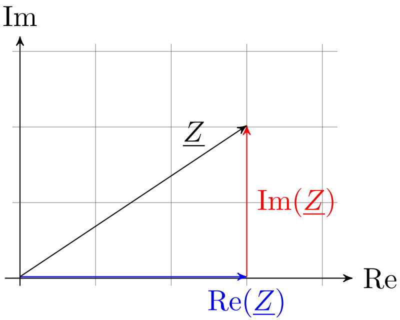
- The real part Re is plotted on the vertical axis and the imaginary part Im on the horizontal axis.



Representation of complex numbers

Use of the complex layer:

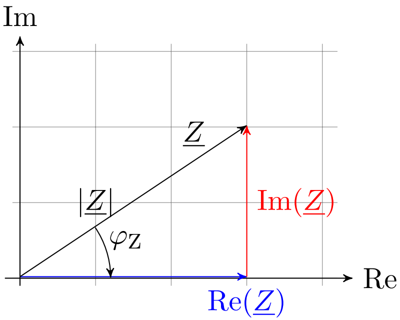
- The real part Re is plotted on the vertical axis and the imaginary part Im on the horizontal axis.
- Cartesian coordinates: $\mathrm{Re}(\underline{Z}) + \mathrm{j} \cdot \mathrm{Im}(\underline{Z})$



Representation of complex numbers

Use of the complex layer:

- The real part Re is plotted on the vertical axis and the imaginary part Im on the horizontal axis.
- Cartesian coordinates: $\mathrm{Re}(\underline{Z}) + \mathrm{j} \cdot \mathrm{Im}(\underline{Z})$
- Polar coordinates: $|\underline{Z}| \cdot e^{j \cdot \varphi_{\mathrm{Z}}}$



Coordinate transformation

- Euler's formula:

$$\mathrm{e}^{\mathrm{j}\cdot\chi} = \cos(\chi) + \mathrm{j}\cdot\sin(\chi)$$

- Cartesian coordinates:

$$\underline{Z} = |\underline{Z}|\cdot\cos(\varphi) + \mathrm{j}\cdot|\underline{Z}|\cdot\sin(\varphi) = \mathrm{Re}(\underline{Z}) + \mathrm{j}\cdot\mathrm{Im}(\underline{Z})$$

- Polar coordinates:

$$\underline{Z} = \sqrt{(\mathrm{Re}(\underline{Z}))^2 + (\mathrm{Im}(\underline{Z}))^2}\cdot\mathrm{e}^{\mathrm{j}\cdot\arctan\left(\frac{\mathrm{Im}(\underline{Z})}{\mathrm{Re}(\underline{Z})}\right)} = |\underline{Z}|\cdot e^{j\cdot\varphi}$$

Representations of complex numbers

Key point: Representations of complex numbers

Cartesian representation:

$$\underline{Z} = \mathrm{Re}(\underline{Z}) + j \cdot \mathrm{Im}(\underline{Z})$$

Polar form:

$$\underline{Z} = |\underline{Z}| \cdot e^{j \cdot \varphi}$$

Trigonometric representation:

$$\underline{Z} = |\underline{Z}|(\cos(\varphi) + j \cdot \sin(\varphi))$$

Basic arithmetic and operations

Conjugation:

$$\underline{Z}^* = (\mathrm{Re}(\underline{Z}) + \mathrm{j} \cdot \mathrm{Im}(\underline{Z}))^* = \mathrm{Re}(\underline{Z}) - \mathrm{j} \cdot \mathrm{Im}(\underline{Z})$$

Conjugation:

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Amount and amount squared

$$|\underline{Z}| = \sqrt{\underline{Z} \cdot \underline{Z}^*} \rightarrow |\underline{Z}|^2 = \underline{Z} \cdot \underline{Z}^*$$

Basic arithmetic and operations

Addition and subtraction:

$$\underline{Z}_1 + \underline{Z}_2 = \mathrm{Re}(\underline{Z}_1) + \mathrm{Re}(\underline{Z}_2) + j \cdot (\mathrm{Im}(\underline{Z}_1) + \mathrm{Im}(\underline{Z}_2))$$
$$\underline{Z}_1 - \underline{Z}_2 = \mathrm{Re}(\underline{Z}_1) - \mathrm{Re}(\underline{Z}_2) - j \cdot (\mathrm{Im}(\underline{Z}_1) - \mathrm{Im}(\underline{Z}_2))$$

Addition and subtraction:

$$\underline{Z}_1 + \underline{Z}_2 = \mathrm{Re}(\underline{Z}_1) + \mathrm{Re}(\underline{Z}_2) + j \cdot (\mathrm{Im}(\underline{Z}_1) + \mathrm{Im}(\underline{Z}_2))$$
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Multiplication and division:

$$\underline{Z}_1 \cdot \underline{Z}_2 = |\underline{Z}_1| \cdot |\underline{Z}_2| \cdot e^{j \cdot (\varphi_1 + \varphi_2)}$$

$$\frac{\underline{Z}_1}{\underline{Z}_2} = \frac{|\underline{Z}_1|}{|\underline{Z}_2|} \cdot e^{j \cdot (\varphi_1 - \varphi_2)}$$

Key point: Basic arithmetic operations with complex numbers

For **addition and subtraction**, the **Cartesian representation** is generally used for complex numbers. For **multiplication and division** of complex numbers, the **polar form** is chosen.

Division by conjugate complex extension:

$$\frac{\underline{Z}_1}{\underline{Z}_2} = \frac{\underline{Z}_1}{\underline{Z}_2} \cdot \frac{\underline{Z}_2^*}{\underline{Z}_2^*} = \frac{\underline{Z}_1 \cdot \underline{Z}_2^*}{|\underline{Z}_2|^*}$$

Basic arithmetic and operations

Division by conjugate complex extension:

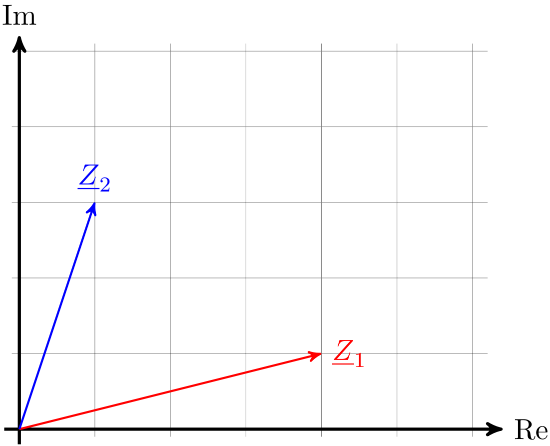
$$\frac{\underline{Z}_1}{\underline{Z}_2} = \frac{\underline{Z}_1}{\underline{Z}_2} \cdot \frac{\underline{Z}_2^*}{\underline{Z}_2^*} = \frac{\underline{Z}_1 \cdot \underline{Z}_2^*}{|\underline{Z}_2|^*}$$

Calculate the real part and imaginary part:

$$\mathrm{Re}(\underline{Z}) = \frac{\underline{Z} + \underline{Z}^*}{2}$$

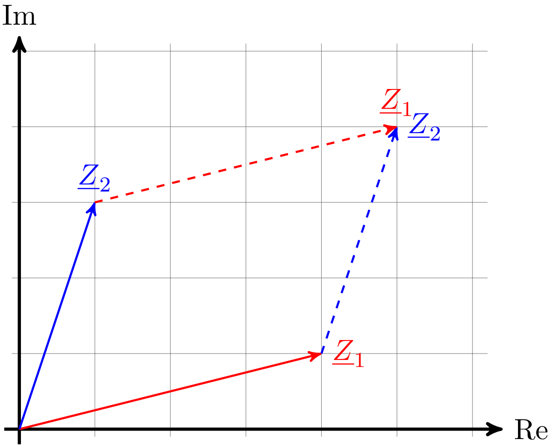
$$\mathrm{Im}(\underline{Z}) = \frac{\underline{Z} - \underline{Z}^*}{2j}$$

Arithmetic operations in the pointer diagram - Addition:



Arithmetic operations in the pointer diagram - Addition:

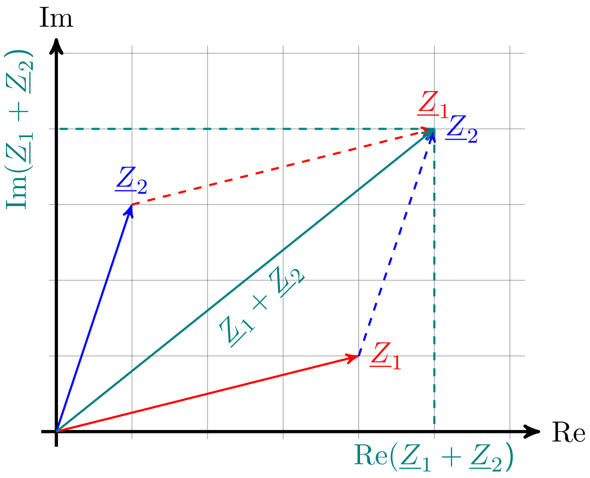
1. Parallel translation Z_2 to the tip of Z_1



Graphical representation and calculations with complex numbers

Arithmetic operations in the pointer diagram - Addition:

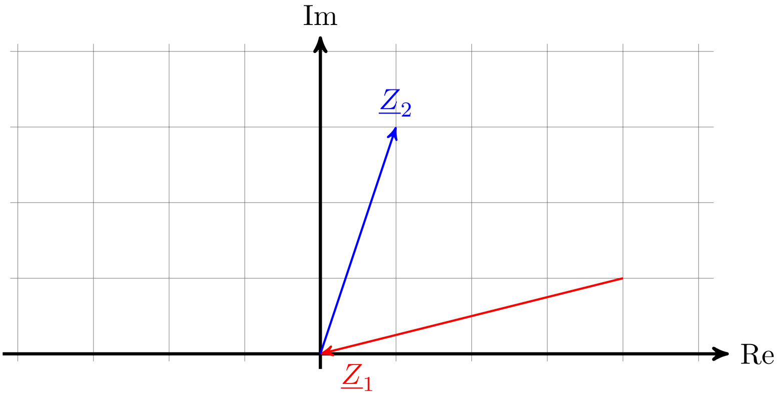
1. Parallel translation Z_2 to the tip of Z_1
2. Read off the real part and imaginary part



Pointer diagrams - Subtraction

Arithmetic operations in the pointer diagram - Subtraction:

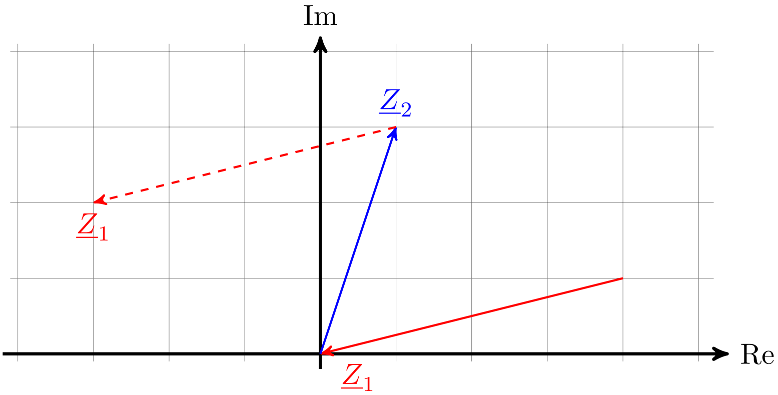
1. Pointer reversal for negation



Pointer diagrams - Subtraction

Arithmetic operations in the pointer diagram - Subtraction:

1. Pointer reversal for negation
2. Move pointer Z_1: Base point of Z_1 to tip of Z_2



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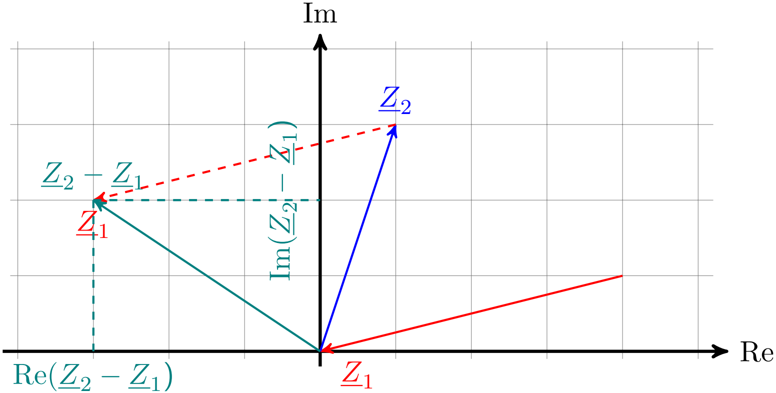
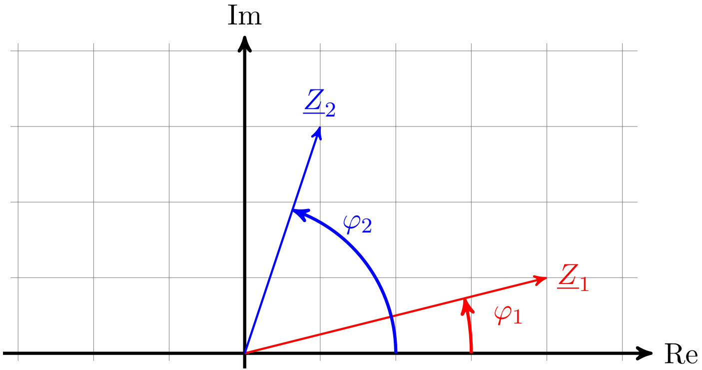


Pointer diagrams - Multiplication

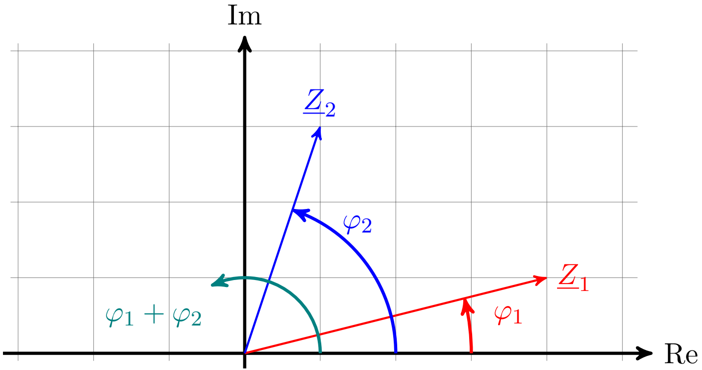
Arithmetic operations in the pointer diagram - Multiplication: (Drawing not to scale!)



Pointer diagrams - Multiplication

Arithmetic operations in the pointer diagram - Multiplication: (Drawing not to scale!)

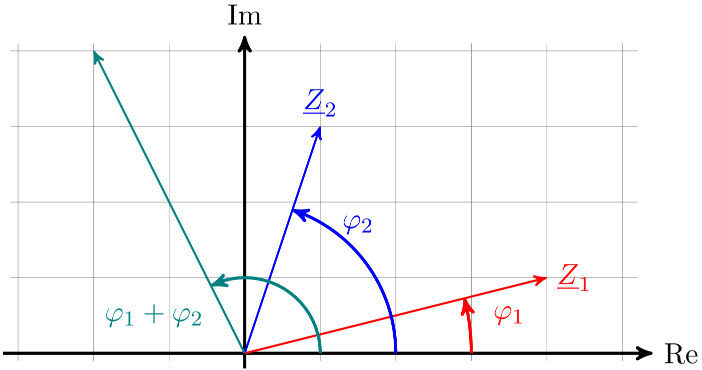
1. Winkel $\varphi_1 + \varphi_2$ berechnen und einzeichnen



Pointer diagrams - Multiplication

Arithmetic operations in the pointer diagram - Multiplication: (Drawing not to scale!)

1. Winkel $\varphi_1 + \varphi_2$ berechnen und einzeichnen
2. Calculate and enter the length $Z_1 \cdot Z_2$



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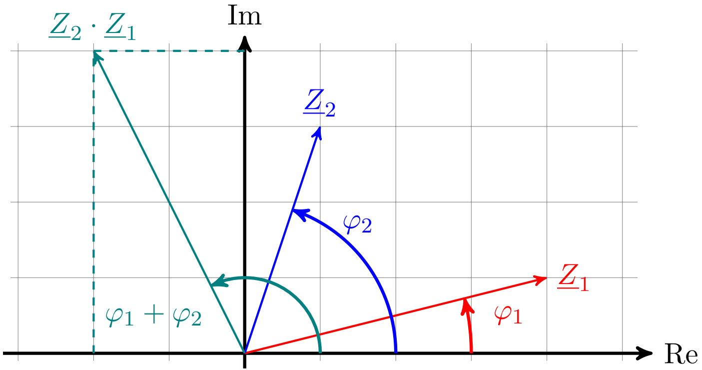


Learning objectives: Pointer diagrams

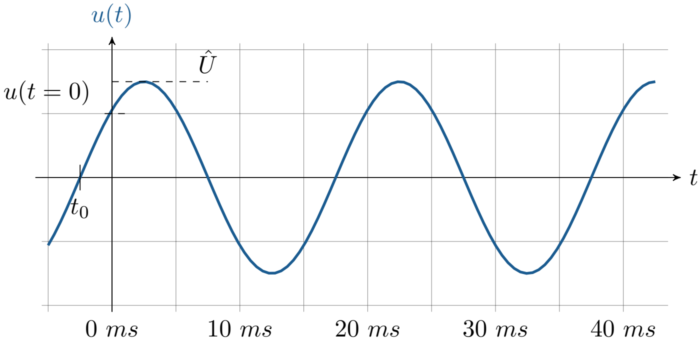
the students can

- understand the parameters of periodic alternating voltages.
- can handle the complex rotation indicators of the amplitudes.

Periodic alternating voltage

Application of complex numbers in alternating current technology

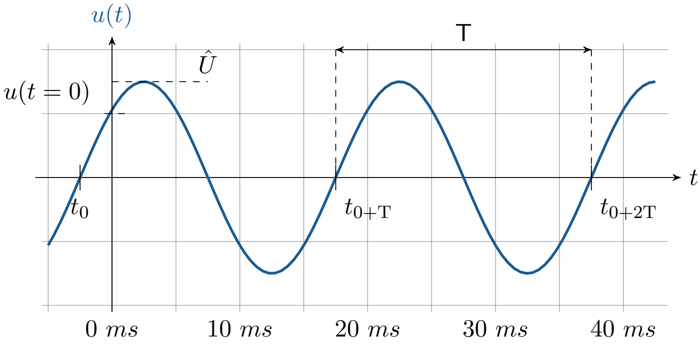
- Sinusoidal and harmonic source magnitudes, peak value $\hat{U}$ and phase shift t_0



Periodic alternating voltage

Application of complex numbers in alternating current technology

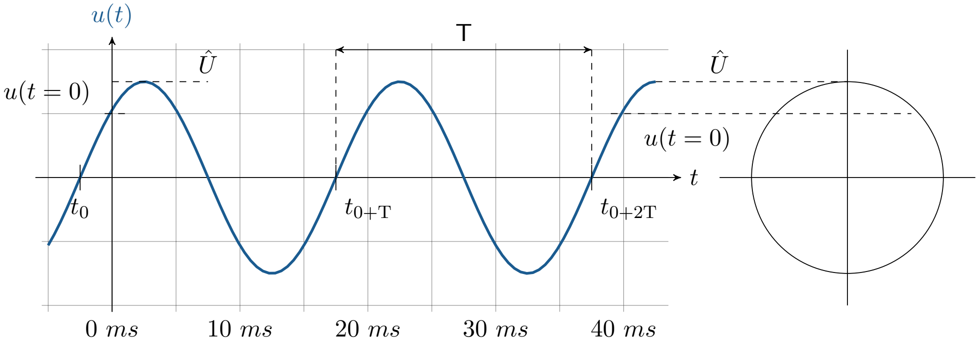
- Sinusoidal and harmonic source magnitudes, peak value $\hat{U}$ and phase shift t_0
- Period duration T, repetition of periods



Periodic alternating voltage

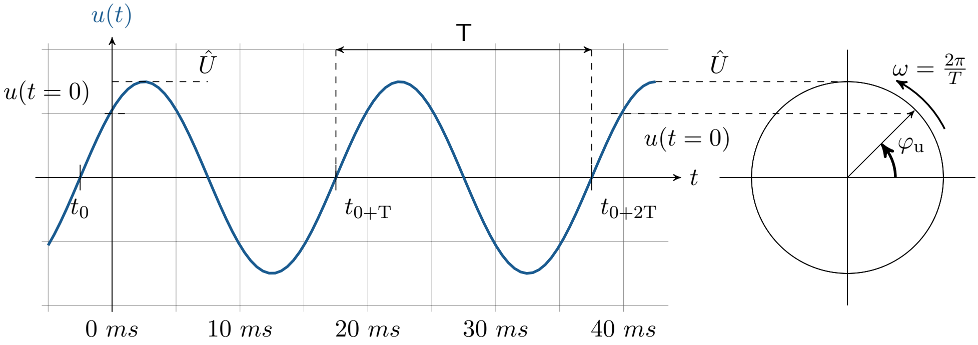
Application of complex numbers in alternating current technology

- Sinusoidal and harmonic source magnitudes, peak value $\hat{U}$ and phase shift t_0
- Period duration T, repetition of periods
- Steady state $\rightarrow$ Voltages and currents differ only in amplitude and phase



Application of complex numbers in alternating current technology

- Sinusoidal and harmonic source magnitudes, peak value $\hat{U}$ and phase shift t_0
- Period duration T, repetition of periods
- Steady state $\rightarrow$ Voltages and currents differ only in amplitude and phase
- Vibrations can be represented by pointers.



Conversion of angular frequency

- Circular frequency ω described by the period T

$$\omega = \frac{2\pi}{T}$$

Conversion of angular frequency

- Circular frequency ω described by the period T
- Conversion of period duration and frequency f

$$\omega = \frac{2\pi}{T} \quad \text{with} \quad f = \frac{1}{T}$$

Conversion of angular frequency

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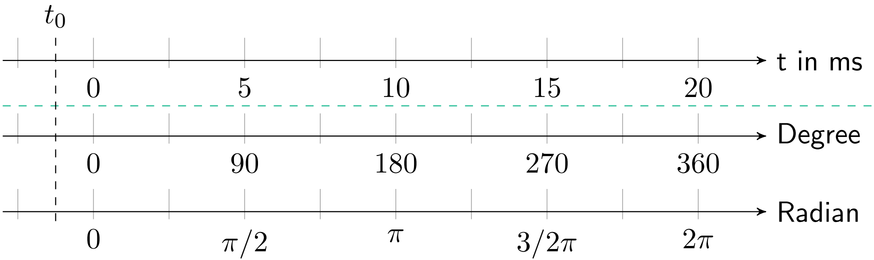
- Circular frequency ω described by the frequency

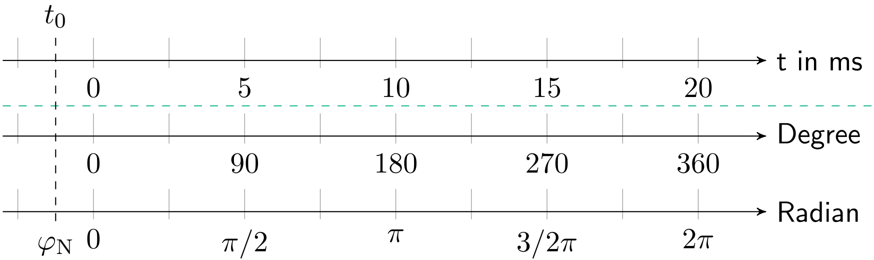
$$\omega = \frac{2\pi}{T} \quad \text{with} \quad f = \frac{1}{T} \quad \text{will be} \quad \omega = 2\pi f$$

Harmonic alternating quantities

- In angular terms, one period corresponds to a circular arc of 360 or 2π.



- In angular terms, one period corresponds to a circular arc of 360 or 2π.
- The phase shift t_0 corresponds to the phase angle in the angular representation. φ_{N}



Alternating quantities

Key point: Exchangeable sizes

An alternating voltage explains a regular polarity change with the amplitude value $\hat{U}$ and the **Periodendauer** T.

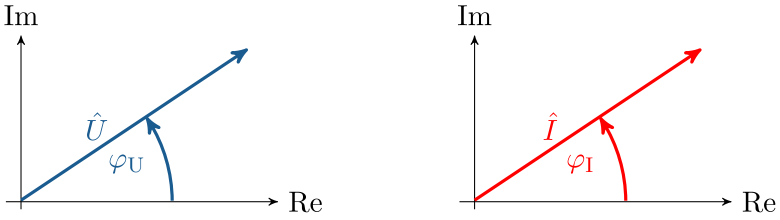
Time-dependent alternating voltage

- $u(t)$ described by the angular frequency ω
- $u(t)$ at time t with a period duration of T

$$u(t) = \hat{U} \cdot \sin(\omega t + \varphi_{\mathrm{N}}) = \hat{U} \cdot \sin(2\pi f t + \varphi_{\mathrm{N}}) = \hat{U} \cdot \sin(2\pi \frac{t}{T} + \varphi_N)$$

Complex rotating pointer of the amplitude

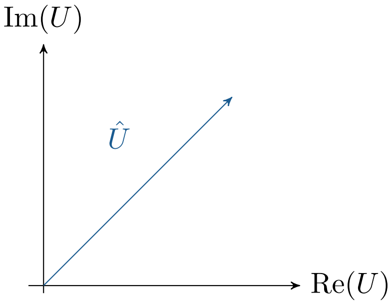
- linear networks $\rightarrow$ sinusoidal and harmonic source quantities
- steady state $\rightarrow$ voltages and currents differ only in amplitude and phase
- oscillations can be represented by phasors
- stationary phasors express the relative position of oscillations to one another



Complex rotating pointer of the amplitude

Complex rotating pointer for electrical voltage:

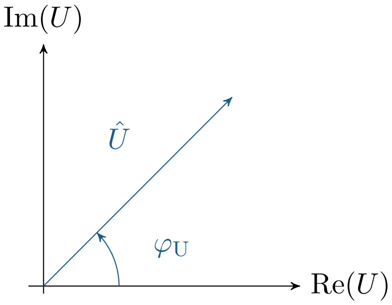
- Amplitude value of the voltage: $\hat{U}$



Complex rotating pointer of the amplitude

Complex rotating pointer for electrical voltage:

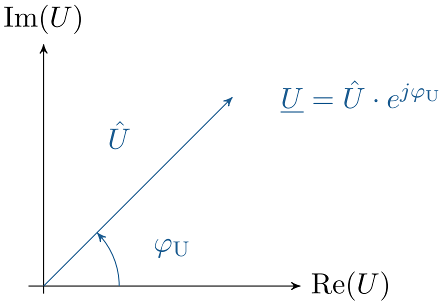
- Amplitude value of the voltage: $\hat{U}$
- Phase angle of the voltage: φ_{U}



Complex rotating pointer of the amplitude

Complex rotating pointer for electrical voltage:

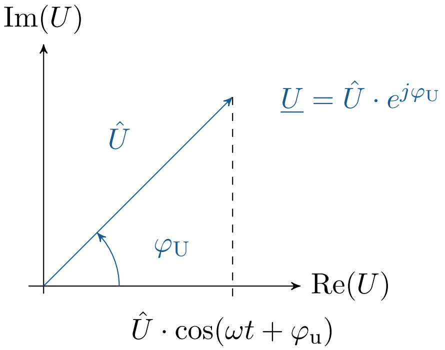
- Amplitude value of the voltage: $\hat{U}$
- Phase angle of the voltage: φ_{U}
- Complex voltage phasor: $\underline{U} = \hat{U} \cdot e^{j\varphi_{\mathrm{U}}}$



Complex rotating pointer of the amplitude

Complex rotating pointer for electrical voltage:

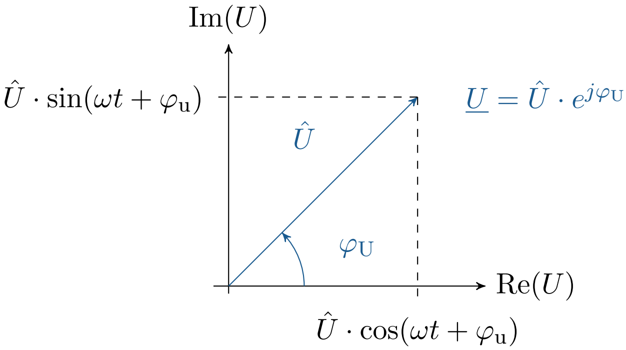
- Amplitude value of the voltage: $\hat{U}$
- Phase angle of the voltage: φ_{U}
- Complex voltage phasor: $\underline{U} = \hat{U} \cdot e^{j\varphi_{\mathrm{U}}}$
- Real part of the voltage



Complex rotating pointer of the amplitude

Complex rotating pointer for electrical voltage:

- Amplitude value of the voltage: $\hat{U}$
- Phase angle of the voltage: φ_{U}
- Complex voltage phasor: $\underline{U} = \hat{U} \cdot e^{j\varphi_{\mathrm{U}}}$
- Real part of the voltage
- Imaginary part of the voltage



Complex rotating pointers for voltage and current

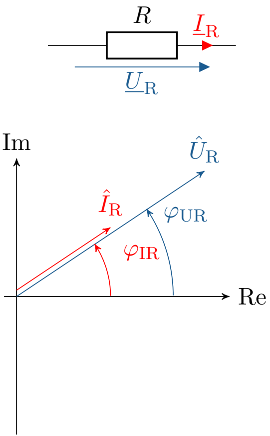
Key point: Complex rotating pointers for voltage and current

The alternating variables voltage and current are represented as complex rotating pointers. The rotating pointers describe **instantaneous values** $u(t)$, which, with constant **angular velocity** ω change.

Voltage indicators and current indicators for electrical components

Voltage indicators and current indicators for various components:

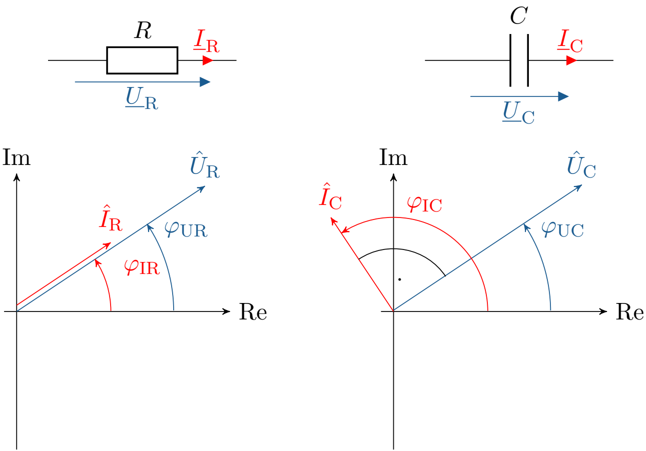
- Electrical resistance $\rightarrow$ voltage phasor and current phasor in phase



Voltage indicators and current indicators for electrical components

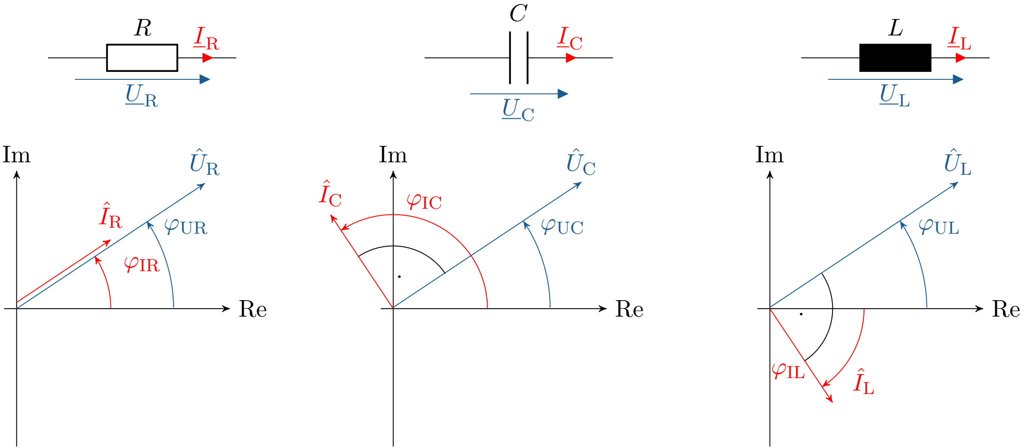
Voltage indicators and current indicators for various components:

- Electrical resistance $\rightarrow$ voltage phasor and current phasor in phase
- Capacitance $\rightarrow$ current phasor leads the voltage phasor



Voltage indicators and current indicators for various components:

- Electrical resistance → voltage phasor and current phasor in phase
- Capacitance → current phasor leads the voltage phasor
- Inductance → current phasor lags behind the voltage phasor



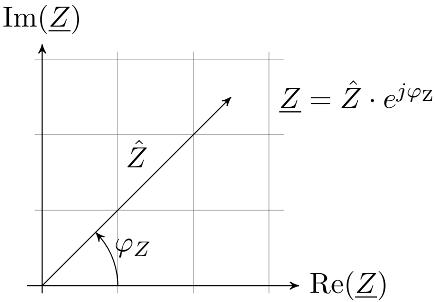
Learning objectives: Alternating current calculation

The students

- know the complex impedance and complex admittance.
- understand the behaviour of a resistor, a capacitor and an inductor at an alternating voltage.
- know alternating current sources and the associated conversion rules.

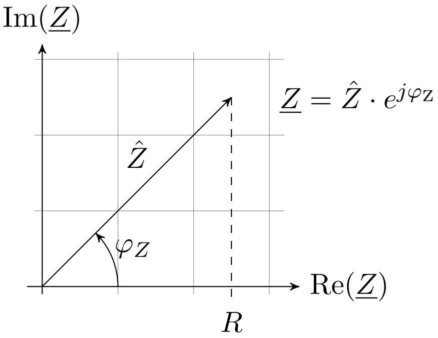
Impedance and admittance in the complex plane

- Complex pointer of impedance



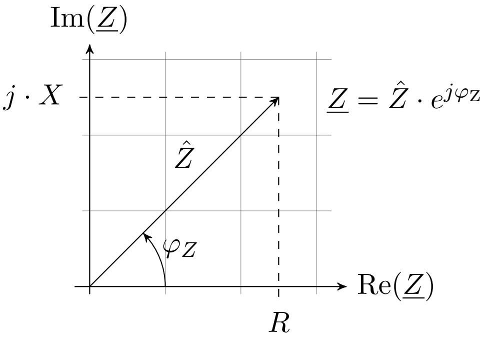
Impedance and admittance in the complex plane

- Complex pointer of impedance
- Active component of impedance

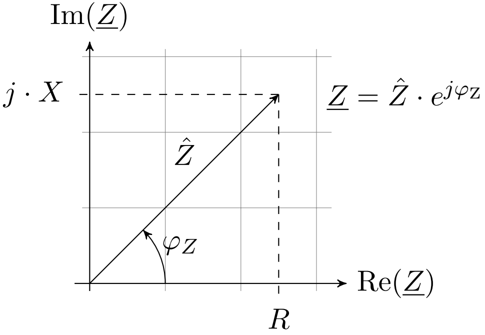


Impedance and admittance in the complex plane

- Complex pointer of impedance
- Active component of impedance
- Reactive component of impedance



- Complex pointer of impedance
- Active component of impedance
- Reactive component of impedance
- Active resistance and reactive resistance of impedance



$$Impedance = Resistance + j \cdot Reactance \quad \rightarrow \quad \underline{Z} = R + j \cdot X$$

$$[\underline{Z}] = 1\ Ohm = 1\ \Omega$$

Impedance and admittance in the complex plane

Complex conductance as the reciprocal of complex impedance:

$$R = \frac{1}{G} \quad \rightarrow \quad \underline{Z} = \frac{1}{\underline{Y}}$$

Impedance and admittance in the complex plane

Complex conductance as the reciprocal of complex impedance:

$$R = \frac{1}{G} \quad \rightarrow \quad \underline{Z} = \frac{1}{\underline{Y}}$$

Effective conductance and reactive conductance of admittance:

$$Admittance = Conductance + j \cdot Susceptibility \quad \rightarrow \quad \underline{Y} = G + j \cdot B$$

$$[\underline{Y}] = 1\ Siemens = 1\ S$$

Complex impedance and complex admittance

Key point: Complex impedance and complex admittance

The direct current resistance becomes the complex impedance $\underline{Z}$ for an alternating voltage. The reciprocal of the resistance, the conductance, becomes the complex admittance $\underline{Y}$ for an alternating voltage.

Complex resistance

- Complex resistance at an ohmic load

$$\underline{Z}_{\mathrm{R}} = R + j \cdot X_{\mathrm{R}}$$

Complex resistance

- Complex resistance at an ohmic load
- No reactive component in an ideal ohmic resistance

$$\underline{Z}_{\mathrm{R}} = R + j \cdot X_{\mathrm{R}} \quad \rightarrow \quad X_{\mathrm{R-ideal}} = j \cdot 0\ \Omega$$

Complex resistance

- Complex resistance at an ohmic load
- No reactive component in an ideal ohmic resistance
- Ideal complex resistance as pure active component

$$\underline{Z}_{\mathrm{R}} = R + j \cdot X_{\mathrm{R}} \quad \rightarrow \quad X_{\mathrm{R-ideal}} = j \cdot 0\ \Omega \quad \rightarrow \quad \underline{Z}_{\mathrm{R}} = R$$

Complex resistance

- Complex resistance at an ohmic load
- No reactive component in an ideal ohmic resistance
- Ideal complex resistance as pure active component

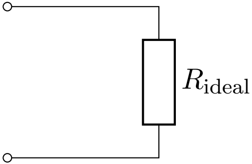
$$\underline{Z}_{\mathrm{R}} = R + j \cdot X_{\mathrm{R}} \quad \rightarrow \quad X_{\mathrm{R-ideal}} = j \cdot 0\ \Omega \quad \rightarrow \quad \underline{Z}_{\mathrm{R}} = R$$

$$\underline{U} = \underline{Z}_{\mathrm{R}} \cdot \underline{I} \quad \rightarrow \quad u(t) = R \cdot i(t)$$

Complex resistance

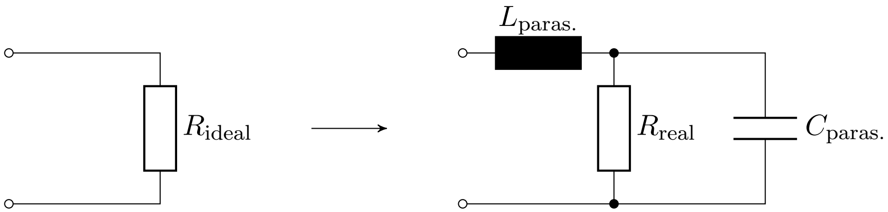
Electrical resistance in alternating current calculations

- Ideal resistance



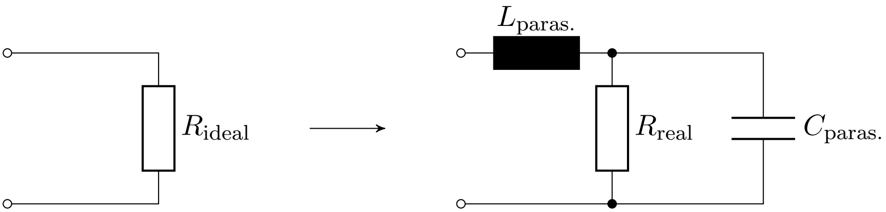
Electrical resistance in alternating current calculations

- Ideal resistance
- Real resistance



Electrical resistance in alternating current calculations

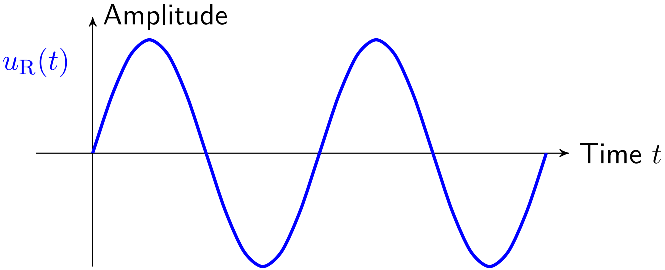
- Ideal resistance
- Real resistance
- No phase shift with ideal resistance



Complex resistance

Phase shift at ideal ohmic resistance:

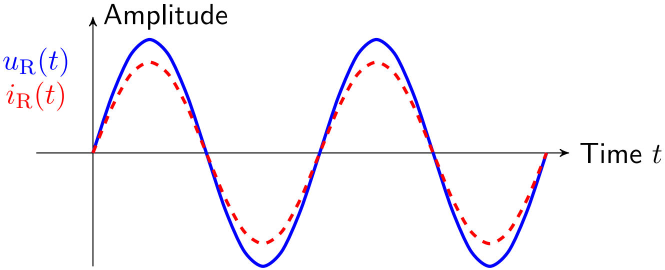
- Voltage $u_{\mathrm{R}}(t)$



Complex resistance

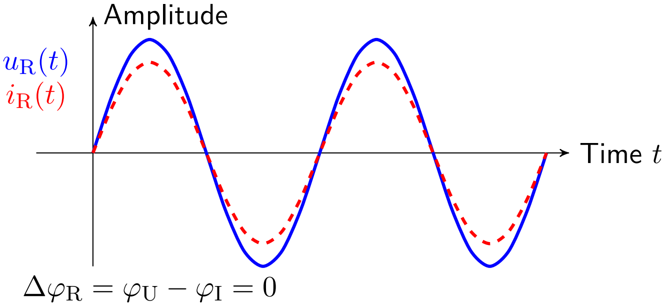
Phase shift at ideal ohmic resistance:

- Voltage $u_\mathrm{R}(t)$
- Current $i_\mathrm{R}(t)$



Phase shift at ideal ohmic resistance:

- Voltage $u_{\mathrm{R}}(t)$
- Current $i_{\mathrm{R}}(t)$
- Voltage and current at the ideal ohmic resistance without phase shift



Key point: Sinusoidal oscillation at a resistor

The **electrical resistance** indicates between the alternating voltage and the corresponding alternating current **no phase shift**.

Capacity

Negative reactive resistance $\leftrightarrow$ Representation in the negative imaginary domain

$$\underline{Z}_{\mathrm{C}} = \frac{1}{j\omega C} \quad \rightarrow \quad \underline{Y}_{\mathrm{C}} = j\omega C$$

Capacity

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Ideal capacitor for alternating voltage and alternating current:

$$\underline{U} = \underline{Z}_\mathrm{C} \cdot \underline{I} \qquad\qquad u(t) = \underline{X}_\mathrm{C} \cdot i(t)$$

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$$\underline{U} = \underline{Z}_{\mathrm{C}} \cdot \underline{I} \qquad\qquad u(t) = \underline{X}_{\mathrm{C}} \cdot i(t)$$

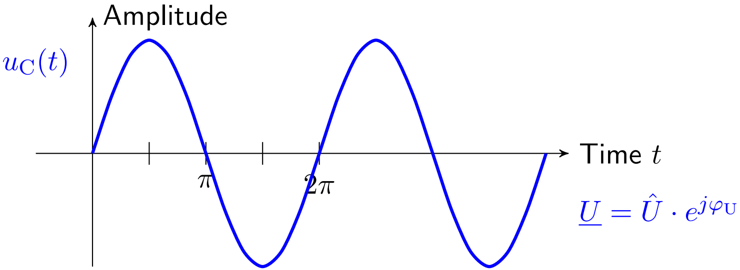
Phase shift of the current at the ideal capacitor:

$$u(t) = \hat{U} \cdot \sin(\omega t) \qquad i(t) = \hat{I} \cdot \cos(\omega t) = \hat{I} \cdot \sin(\omega t + \pi/2)$$

Capacity

Phase shift at the capacitor:

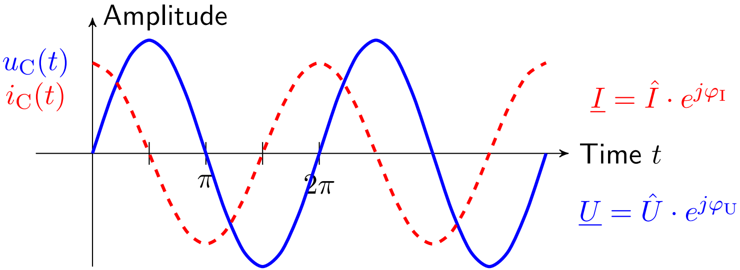
- Voltage $u_{\mathrm{C}}(t)$



Capacity

Phase shift at the capacitor:

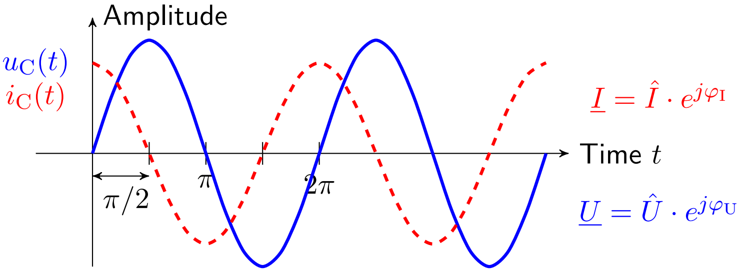
- Voltage $u_\mathrm{C}(t)$
- Current $i_\mathrm{C}(t)$



Capacity

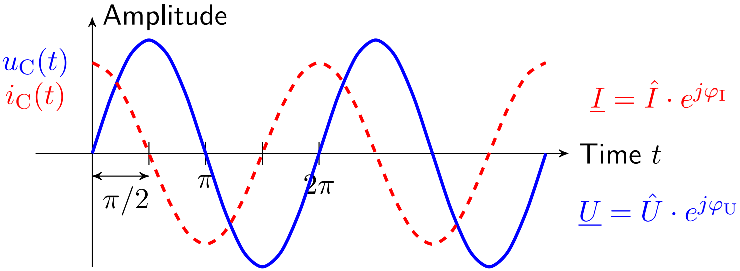
Phase shift at the capacitor:

- Voltage $u_{\mathrm{C}}(t)$
- Current $i_{\mathrm{C}}(t)$
- Phase shift between current and voltage



Phase shift at the capacitor:

- Voltage $u_\mathrm{C}(t)$
- Current $i_\mathrm{C}(t)$
- Phase shift between current and voltage
- Phase shift $\Delta\varphi$



$$\Delta\varphi_\mathrm{C} = \varphi_\mathrm{U} - \varphi_\mathrm{I} = -\pi/2$$

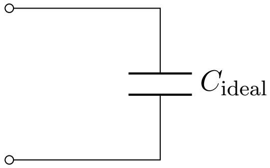
Key point: Sinusoidal oscillation on a capacitor

At the ideal capacity, the alternating current of the alternating voltage leads by 90°.

Ideal and real capacitor

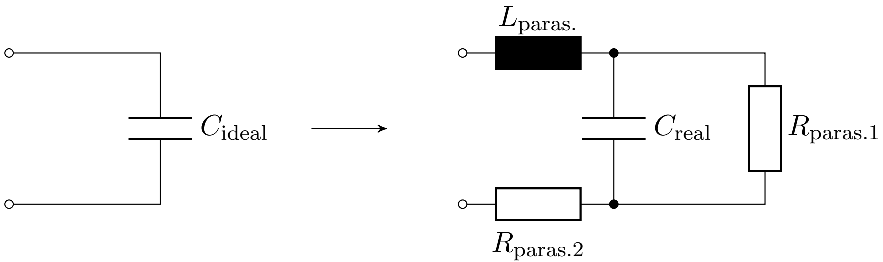
Ideal and real capacitor:

- Ideal capacitor



Ideal and real capacitor:

- Ideal capacitor
- Real capacitor with parasitic effects



Inductance

Positive reactance $\leftrightarrow$ Representation in the positive imaginary domain

$$\underline{Z}_{\mathrm{L}} = j\omega L \quad \rightarrow \quad \underline{Y}_{\mathrm{L}} = \frac{1}{j\omega L}$$

Ideal coil for alternating voltage and alternating current:

$$\underline{U} = \underline{Z}_{\mathrm{L}} \cdot \underline{I} \qquad\qquad u(t) = \underline{X}_{\mathrm{L}} \cdot i(t)$$

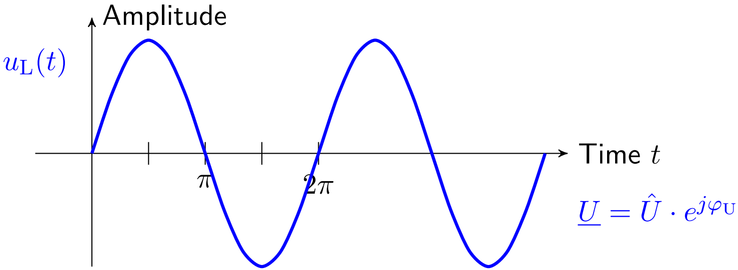
Phase shift of the current at the ideal coil:

$$u(t) = \hat{U} \cdot \cos(\omega t) = \hat{U} \cdot \sin(\omega t + \pi/2) \qquad i(t) = \hat{I} \cdot \sin(\omega t)$$

Inductance

Phase shift at the coil:

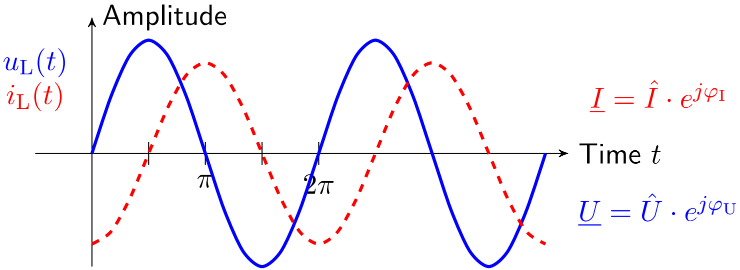
- Voltage $u_{\mathrm{L}}(t)$



Inductance

Phase shift at the coil:

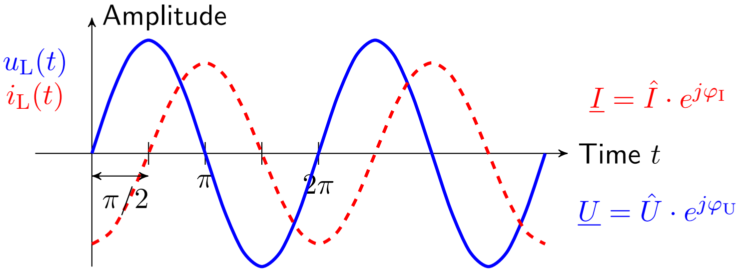
- Voltage $u_{\mathrm{L}}(t)$
- Current $i_{\mathrm{L}}(t)$



Inductance

Phase shift at the coil:

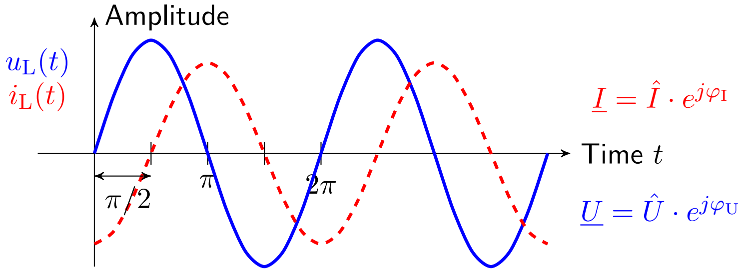
- Voltage $u_{\mathrm{L}}(t)$
- Current $i_{\mathrm{L}}(t)$
- Phase shift between current and voltage



Inductance

Phase shift at the coil:

- Voltage $u_\mathrm{L}(t)$
- Current $i_\mathrm{L}(t)$
- Phase shift between current and voltage
- Phase shift $\Delta\varphi$



$$\Delta\varphi_\mathrm{L} = \varphi_\mathrm{U} - \varphi_\mathrm{I} = +\pi/2$$

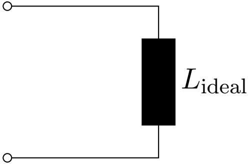
Key point: Sinusoidal oscillation on a coil

With ideal inductance, the alternating current lags the alternating voltage by 90°.

Ideale und reale Spule

Ideal and real coil:

- Ideal coil



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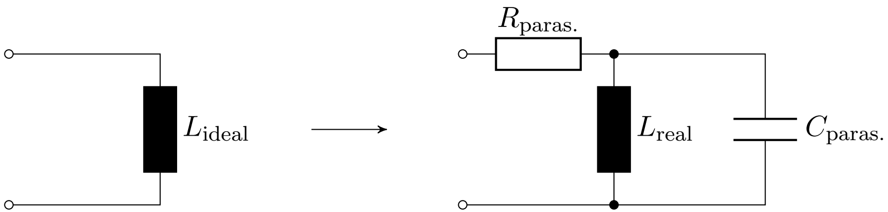
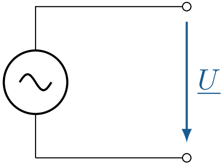
- Real coil with parasitic effects



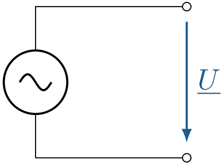
Sources of alternating quantities:

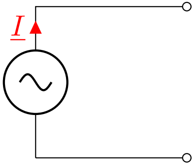
- Ideal alternating voltage source



Sources of alternating quantities:

- Ideal alternating voltage source
- Ideal alternating current source

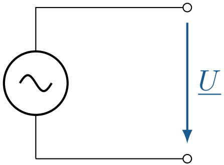




Real AC voltage source

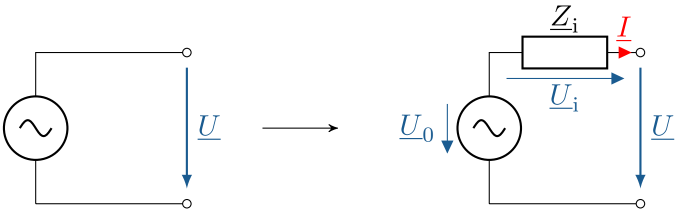
Alternating voltage source:

- Ideal AC voltage source



Alternating voltage source:

- Ideal AC voltage source
- Real AC voltage source with internal impedance $\underline{Z}_\mathrm{i}$



Alternating voltage source:

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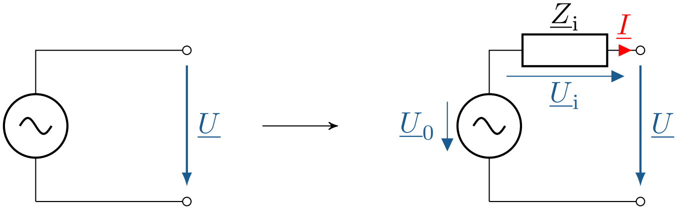


Output voltage of a real AC voltage source:

$$\underline{U} = \underline{U}_0 - \underline{Z}_\mathrm{i} \cdot \underline{I}$$

Characteristics of real AC voltage sources

No-load:

$$\underline{I} = 0 \qquad \rightarrow \qquad \underline{U}_\mathrm{i} = \underline{I} \cdot \underline{Z}_\mathrm{i} = 0$$

$$\underline{U}_0 - \underline{U}_\mathrm{i} - \underline{U} = 0 \qquad \rightarrow \qquad \underline{U} = \underline{U}_0$$

Short circuit:

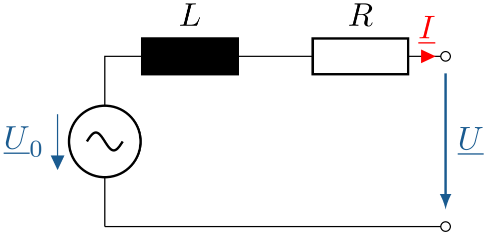
$$\underline{U} = 0 \qquad \rightarrow \qquad \underline{I} = \underline{I}_\mathrm{k}$$

$$\underline{U}_0 - \underline{U}_\mathrm{i} - \underline{U} = 0 \qquad \rightarrow \qquad \underline{U}_\mathrm{i} = \underline{U}_0$$

$$\underline{I}_\mathrm{k} = \frac{\underline{U}_\mathrm{i}}{\underline{Z}_\mathrm{i}}$$

Real voltage source:

A real voltage source (230 V, 50 Hz) has an internal impedance $\underline{Z}_\mathrm{i}$ as shown in the figure below, which consists of a coil $L = 20\ mH$ and an ohmic resistance $R = 10\ m\Omega$.



Determine the **open-circuit voltage** and the **short-circuit current** AC voltage source.

Real voltage source:

$$\underline{U}_0 = 230\ V \cdot e^{\mathrm{j}0^o}$$
$$\underline{Z}_\mathrm{i} = R + jX_L$$
$$\underline{Z}_\mathrm{i} = 0{,}01\ \Omega + j \cdot 2\pi \cdot 50\ Hz \cdot 20\ mH$$
$$\underline{Z}_\mathrm{i} = 0{,}01\ \Omega + j6{,}28\ \Omega$$

Open-circuit voltage:

$$\underline{U} = \underline{U}_0$$
$$\underline{U} = 230\ V \cdot e^{\mathrm{j}0^o}$$

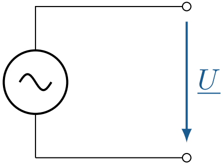
Short-circuit current:

$$\underline{I}_\mathrm{k} = \frac{\underline{U}_0}{\underline{Z}_\mathrm{i}}$$
$$\underline{I}_\mathrm{k} = \frac{230\ V \cdot e^{\mathrm{j}0^o}}{0{,}01\ \Omega + j6{,}28\ \Omega}$$
$$\underline{I}_\mathrm{k} = (0{,}058 - j36{,}6)\ A = 36{,}6\ A \cdot e^{-\mathrm{j}89{,}9^o}$$

Real AC source

AC source:

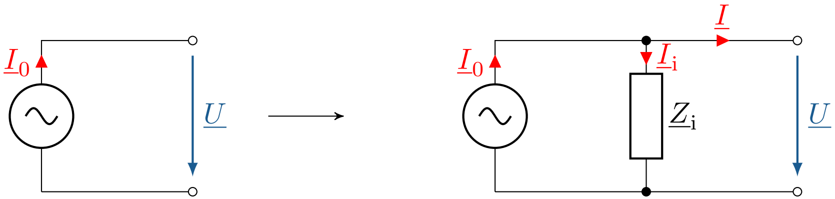
- Ideal AC source



Real AC source

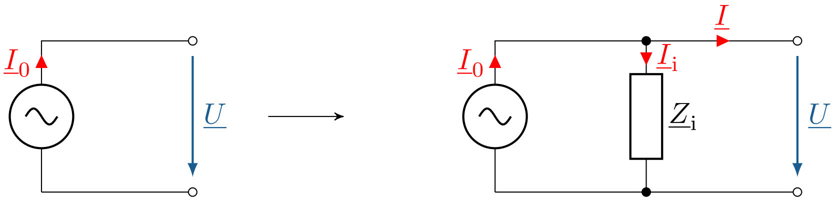
AC source:

- Ideal AC source
- Real AC source with internal impedance $\underline{Z}_\mathrm{i}$



AC source:

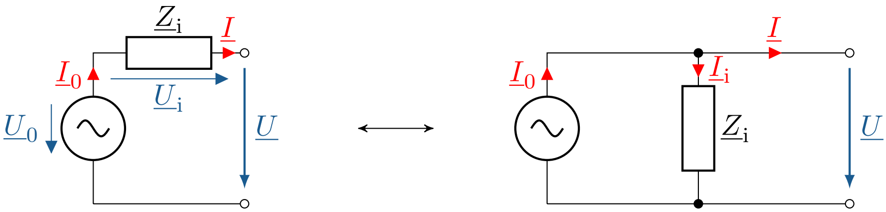
- Ideal AC source
- Real AC source with internal impedance $\underline{Z}_\mathrm{i}$



Output alternating current of a real alternating current source:

$$\underline{I} = \underline{I}_0 - \underline{I}_\mathrm{i} = \underline{I}_0 - \frac{\underline{U}}{\underline{Z}_\mathrm{i}}$$

Conversion of alternating current sources and alternating voltage sources:



Equivalent AC sources, with identical internal impedances:

$$\underline{U}_0 = \underline{Z}_0 \cdot \underline{I}_0$$

$$\underline{Z}_\mathrm{i} = \underline{Z}_0$$

Source current of the current source equals the short-circuit current.
Source voltage of the voltage source equals the open-circuit voltage.

$$\underline{I} = \underline{I}_\mathrm{k} = \underline{I}_0 \qquad \text{und} \qquad \underline{U} = \underline{U}_\mathrm{L} = \underline{U}_0$$

Example task

Alternating current conversion:

Given is a real AC voltage source with $\underline{U}_0 = 20\ kV \cdot \mathrm{e}^{\mathrm{j}30^o}$ and $\underline{Z}_0 = (0.01 + j \cdot 2)\ \Omega$. Convert the alternating voltage source into an equivalent alternating current source and calculate the short-circuit current.

Alternating current conversion:

Short-circuit current:

$$\underline{I}_0 = \frac{\underline{U}_0}{\underline{Z}_0} = \frac{20\ kV \cdot \mathrm{e}^{\mathrm{j}30^o}}{(0{,}01 + j \cdot 2)\ \Omega} = (5{,}04 + j \cdot 8{,}64)\ kA$$

Complex voltage divider and complex current divider

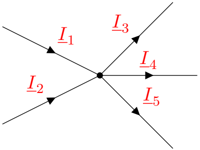
As with the consideration of direct current networks, the basic laws apply to the analysis of alternating current networks.

Complex voltage divider and complex current divider

As with the consideration of direct current networks, the basic laws apply to the analysis of alternating current networks.

- Knot rule:

$$\sum_{k=1}^{N} \underline{I}_{\mathrm{k}} = 0 = \underline{I}_1 + \underline{I}_2 - \underline{I}_3 - \underline{I}_4 - \underline{I}_5$$



Complex voltage divider and complex current divider

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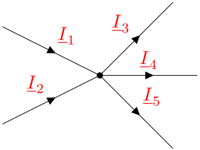
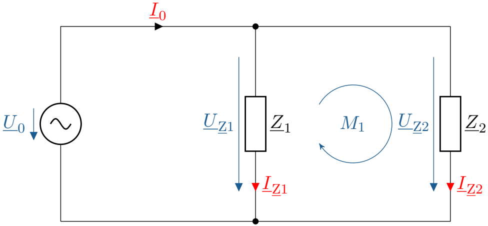
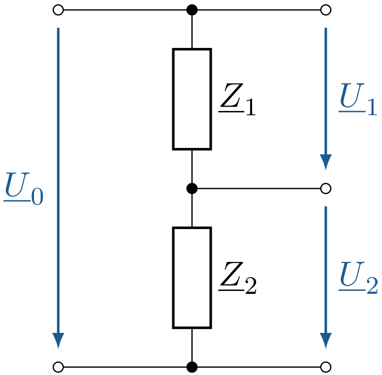
- Mesh rule:

$$\sum_{\mathrm{k}=1}^{N} \underline{U}_{\mathrm{k}} = \underline{U}_{\mathrm{Z2}} - \underline{U}_{\mathrm{Z1}} = 0$$

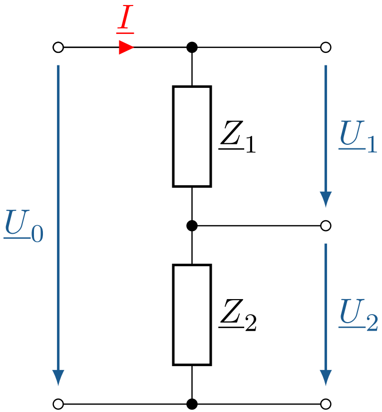




Complex voltage divider

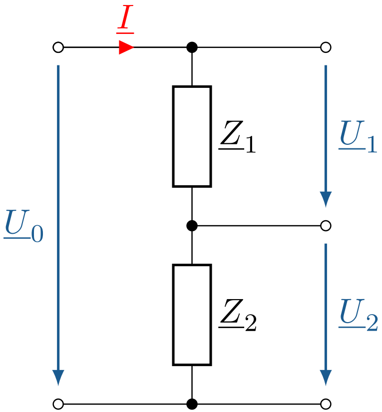


- Reduction of a total voltage, e.g., to adjust a signal level to the input voltage range of a measuring circuit.



- Reduction of a total voltage, e.g., to adjust a signal level to the input voltage range of a measuring circuit.
- The total voltage is divided in proportion to the complex impedances:

$$\underline{I} = \frac{\underline{U}_0}{\underline{Z}_1 + \underline{Z}_2} = \frac{\underline{U}_1}{\underline{Z}_1} = \frac{\underline{U}_2}{\underline{Z}_2}$$



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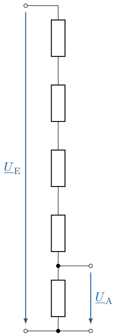
- Complex divisor ratio

$$\underline{T} = \frac{\underline{U}_2}{\underline{U}_0} = \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2}$$

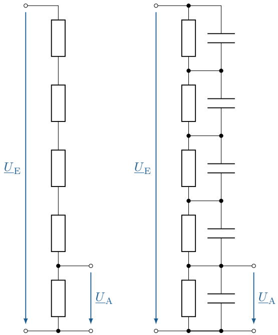
Complex voltage divider: Application example high-voltage technology



Ohmscher Teiler

- Ohmic divider (DC voltage)

Complex voltage divider: Application example high-voltage technology

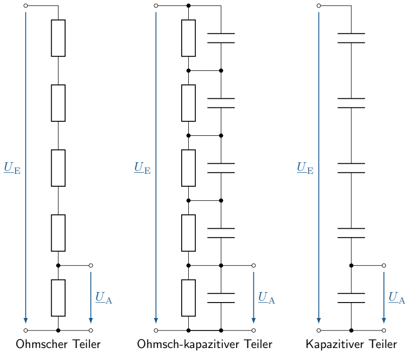


Ohmscher Teiler

Ohmsch-kapazitiver Teiler

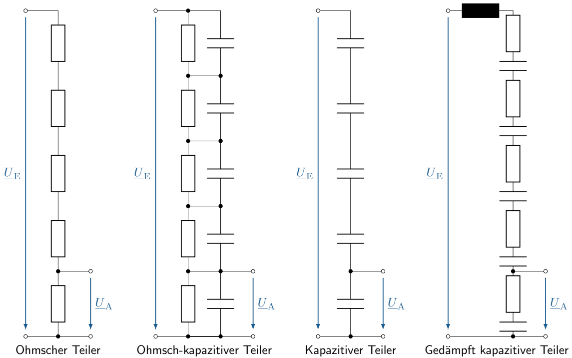
- Ohmic divider (DC voltage)
- Ohmic-capacitive divider (DC and AC voltage)

Complex voltage divider: Application example high-voltage technology



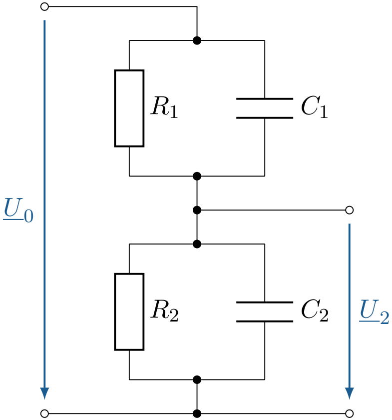
- Ohmic divider (DC voltage)
- Ohmic-capacitive divider (DC and AC voltage)
- Capacitive divider (AC voltage)

Complex voltage divider: Application example high-voltage technology



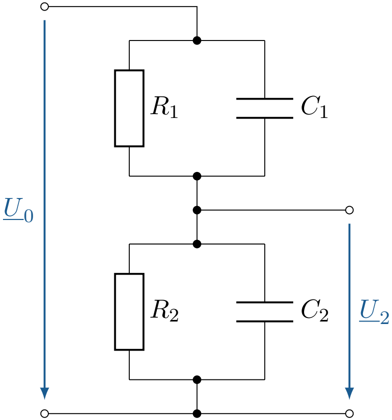
- Ohmic divider (DC voltage)
- Ohmic-capacitive divider (DC and AC voltage)
- Capacitive divider (AC voltage)
- Damped capacitive divider (AC voltage)

Complex voltage divider



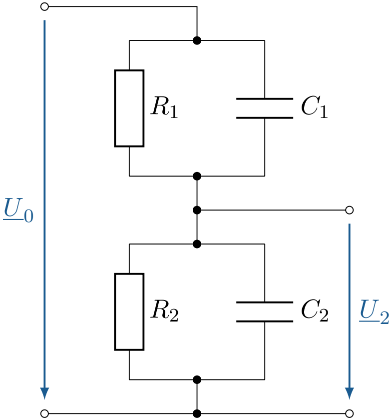
- Ohmic and capacitive division ratio

Complex voltage divider



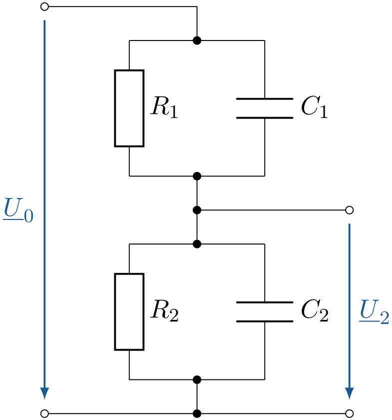
- Ohmic and capacitive division ratio
- Impedances dependent on frequency

Complex voltage divider



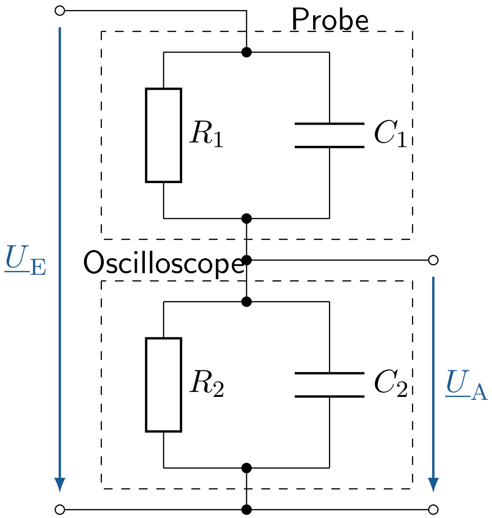
- Ohmic and capacitive division ratio
- Impedances dependent on frequency
- Complex division ratio is frequency-dependent

Complex voltage divider



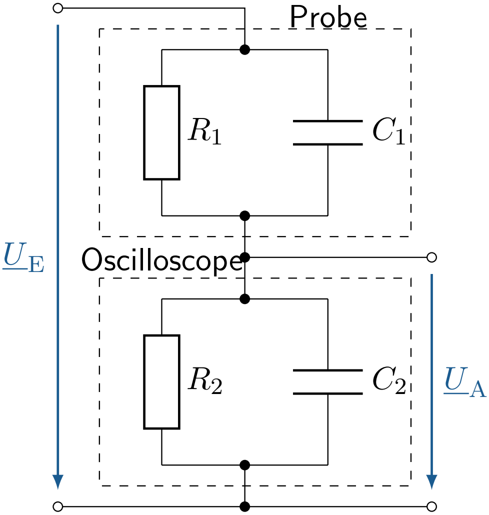
- Ohmic and capacitive division ratio
- Impedances dependent on frequency
- Complex division ratio is frequency-dependent
- The division ratio is only in exceptional cases independent of frequency!

$$R_1 \cdot C_1 = R_2 \cdot C_2$$



- Ohmic ratio:

$$\frac{\underline{U}_\mathrm{A}}{\underline{U}_\mathrm{E}} = \frac{R_2}{R_1 + R_2}$$

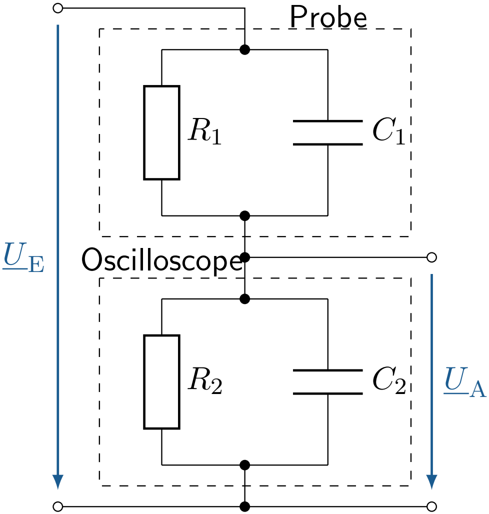


- Ohmic ratio:

$$\frac{\underline{U}_\mathrm{A}}{\underline{U}_\mathrm{E}} = \frac{R_2}{R_1 + R_2}$$

- Capacitive partial ratio:

$$\frac{\underline{U}_\mathrm{A}}{\underline{U}_\mathrm{E}} = \frac{C_2}{C_1 + C_2}$$



- Ohmic ratio:

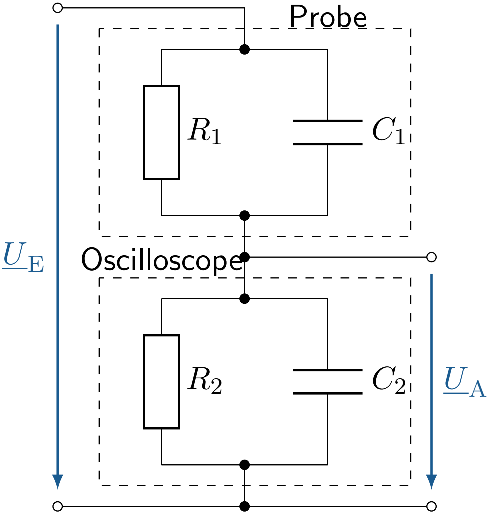
$$\frac{\underline{U}_\mathrm{A}}{\underline{U}_\mathrm{E}} = \frac{R_2}{R_1 + R_2}$$

- Capacitive partial ratio:

$$\frac{\underline{U}_\mathrm{A}}{\underline{U}_\mathrm{E}} = \frac{C_2}{C_1 + C_2}$$

- Equivalence of ohmic and capacitive partial ratios:

$$\frac{R_2}{R_1 + R_2} = \frac{C_2}{C_1 + C_2}$$



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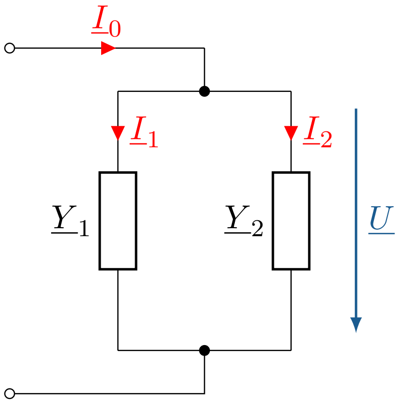
- Equivalence of ohmic and capacitive partial ratios:

$$\frac{R_2}{R_1 + R_2} = \frac{C_2}{C_1 + C_2}$$

- Comparison condition:

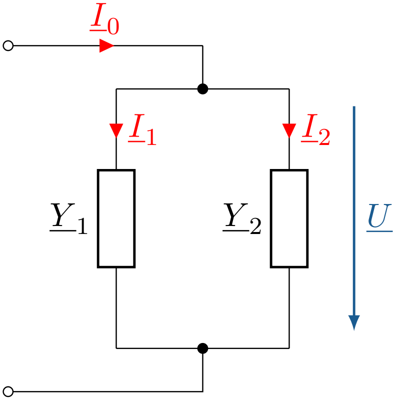
$$R_2 \cdot C_2 = R_1 \cdot C_1$$



- The total current is divided in proportion to the complex admittances:

$$\underline{U} = \frac{\underline{I}_0}{\underline{Y}_1 + \underline{Y}_2} = \frac{\underline{I}_1}{\underline{Y}_1} = \frac{\underline{I}_2}{\underline{Y}_2}$$

Complex current divider



- The total current is divided in proportion to the complex admittances:

$$\underline{U} = \frac{\underline{I}_0}{\underline{Y}_1 + \underline{Y}_2} = \frac{\underline{I}_1}{\underline{Y}_1} = \frac{\underline{I}_2}{\underline{Y}_2}$$

- Impedances depending on frequency

$$\underline{T}_{\mathrm{i}1} = \frac{\underline{I}_1}{\underline{I}_0} = \frac{\underline{Y}_1}{\underline{Y}_1 + \underline{Y}_2}$$
$$\underline{T}_{\mathrm{i}2} = \frac{\underline{I}_2}{\underline{I}_0} = \frac{\underline{Y}_2}{\underline{Y}_1 + \underline{Y}_2}$$

Learning objectives: RMS value

The students

- understand the function of the effective value.
- are familiar with the effects of amplitude and waveform on the effective value.
- can calculate the effective value of an alternating voltage or alternating current.

Fundamentals: Mean square and addition theorem

The effective value enables the comparability of alternating voltages through the root mean square (RMS) value. The RMS is determined using the square root of the mean value and the quadrature of a function:

$$RMS = \sqrt{\frac{1}{T}\int\limits_0^T f(t)^2 dt}$$

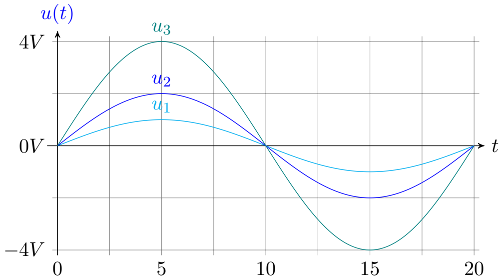
The following addition theorem is helpful when calculating the effective value:

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$$

Amplitude

Relationship between amplitude value and effective value:

- The amplitude describes the peak value $\hat{U}$ of the sinusoidal voltage
- Different amplitudes for different sinusoidal voltages
- The effective value is directly proportional to the amplitude value $U_{\mathrm{RMS}} \sim \hat{U}$



Relationship between the peak value and the effective value via the crest factor:

$$\hat{U} = C \cdot U_{\mathrm{RMS}}$$

Form factors of different waveforms:

Kurvenform:	Sinus	Triangle	Rectangle
Scheitelfaktor (C):	$\sqrt{2}$	$\sqrt{3}$	1

Factors affecting the effective value

Key point: Influences on the RMS value

The RMS value is linearly proportional to the peak value. The conversion factor from the RMS value to the peak value is called the crest factor and depends on the waveform of the signal.

RMS value

$$f(t) = i(t) = \hat{I} \cdot \sin(\omega t) \quad \rightarrow \quad I_{\text{Eff}} = \sqrt{\frac{1}{T} \int_0^T \hat{I}^2 \cdot \sin^2(\omega t) dt}$$

$$I_{\text{RMS}} = \sqrt{\frac{\hat{I}^2}{2T} \int_0^T 1 - \cos(2\omega t) dt}$$

with $\int_0^T 1 dt = T$ and $\int_0^T \cos(2\omega t) dt = 0$

$$I_{\text{RMS}} = \sqrt{\frac{\hat{I}^2}{2T} \cdot (T - 0)}$$

$$I_{\text{RMS}} = \sqrt{\frac{\hat{I}^2}{2}}$$

Effective value of a sinusoidal oscillation

Key point: RMS value of a sinusoidal oscillation

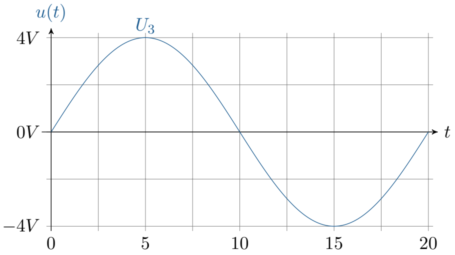
For purely sinusoidal voltages and currents, the crest factor $\sqrt{2}$ applies:

$$I_{\mathrm{RMS}} = \frac{\hat{I}}{\sqrt{2}} \qquad \text{and} \qquad U_{\mathrm{RMS}} = \frac{\hat{U}}{\sqrt{2}}$$

Complex fixed indicator of the effective value

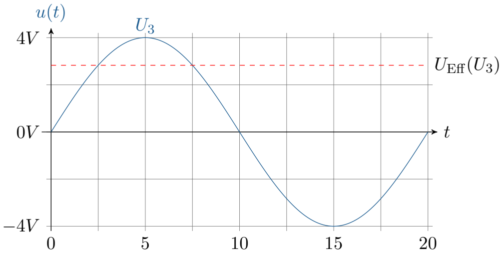
Effective value of an alternating voltage:

- Waveform of a sinusoidal voltage with an amplitude value of $4\,V$



Effective value of an alternating voltage:

- Waveform of a sinusoidal voltage with an amplitude value of $4\,V$
- Corresponding effective value at $4\,V/\sqrt{2}$



Example task

RMS-value:

The following tasks should be completed to gain a better understanding of the RMS value:

a) Calculation of the amplitude value of the voltage at a domestic power socket.
b) Calculation of the RMS value assuming that it is a rectangular signal.

Example task

RMS-value:

a) The amplitude value at a socket with an RMS value of 230V is:

$$\hat{U} = I_{\text{RMS}} \cdot \sqrt{2} = 230V \cdot \sqrt{2}$$
$$\hat{U} = 325{,}269V$$

RMS-value:

b) The RMS value of a square wave voltage with a peak value of 325V is:

$$U_{\text{Rectangle}} = \frac{\hat{U}}{\sqrt{3}} = \frac{325V}{\sqrt{3}}$$
$$U_{\text{Rectangle}} = 187{,}639V$$

Apparent power, active power and reactive power

Learning objectives: Apparent power, active power and reactive power

The students

- know the difference between instantaneous power, active power and reactive power.
- can determine and categorise apparent power.

Addition theorem for converting a product of two sine functions with different arguments:

$$\sin(x) \cdot \sin(y) = \frac{1}{2}(\cos(x-y) + cos(x+y))$$

The basis for calculating the power of alternating quantities is the arithmetic mean value:

- A constant value is to be calculated that is no longer time-dependent
- Serves for better illustration, similar to the effective value

$$\overline{u(t)} = \frac{1}{T} \int\limits_{t}^{t+T} u(t) dt$$

With:

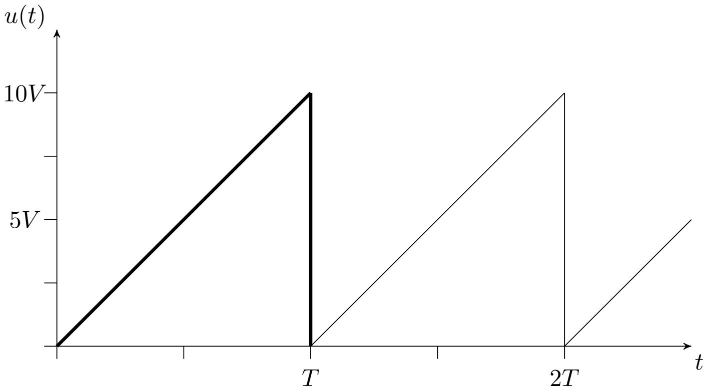
- T=period length
- t=arbitrarily selectable starting point (usually t=0)

Arithmetic mean: Example

Example of averaging based on a sawtooth voltage:

- First, a function is set up over a period length.

$$u(t) = \frac{10\ V}{T} \cdot t$$



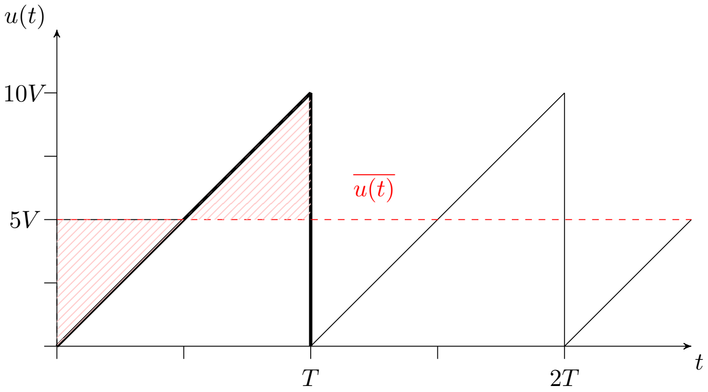
Arithmetic mean: Example

Example of averaging based on a sawtooth voltage:

- First, a function is set up over a period length.

$$u(t) = \frac{10\ V}{T} \cdot t$$

- Arithmetic mean $\overline{u(t)}$:



Arithmetic mean: Example

- This function is used in the equation for the arithmetic mean.

$$\overline{u(t)} = \frac{1}{T} \int_{0}^{T} \frac{10\ V}{T} \cdot t\ dt$$

- After solving the integral, you obtain the arithmetic mean value of the sawtooth voltage.

$$\overline{u(t)} = \frac{1}{T} \left[\frac{10\ V}{T} \cdot \frac{t^2}{2} \right]_0^T = \frac{1}{2} \cdot 10\ V$$

- The arithmetic mean for pure alternating quantities is always zero.

Active power and reactive power

Active power with direct current:

- Power calculation

$$P = U \cdot I = I^2 \cdot R = \frac{U^2}{R}$$

Active power Resistance

- Time-dependent variables:

$$P = U \cdot I \quad \Leftrightarrow \quad p(t) = u(t) \cdot i(t)$$

- Time-dependent voltage and currents:

$$u(t) = \hat{U} \cdot \sin(\omega t + \varphi_{\mathrm{U}}) \qquad i(t) = \hat{I} \cdot \sin(\omega t + \varphi_{\mathrm{I}})$$

- **Instantaneous power**:

$$\text{Generally:} \quad p(t) = u(t) \cdot i(t)$$

$$\text{Sinusoidal:} \quad p(t) = \frac{\hat{U} \cdot \hat{I}}{2} \Big(\underbrace{\cos(\varphi_{\mathrm{U}} - \varphi_{\mathrm{I}})}_{\text{constant proportion}} + \underbrace{\cos(2\omega t + \varphi_{\mathrm{U}} + \varphi_{\mathrm{I}})}_{\text{time-dependent portion}} \Big)$$

Key point: Instantaneous power

The instantaneous power is the product of the instantaneous values of voltage and current.

$$p(t) = u(t) \cdot i(t)$$

Average power

Average power:

- RMS-value of voltage U and current I

$$\overline{p}_{\mathrm{Sin}} = P = U \cdot I \cdot \cos(\varphi)$$

Average power

Average power:

- RMS-value of voltage U and current I
- Phase shift between voltage and current
- **Power factor**: $\cos(\varphi)$

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Average power:

- RMS-value of voltage U and current I
- Phase shift between voltage and current
- **Power factor**: $\cos(\varphi)$

$$\overline{p}_{\text{Sin}} = P = U \cdot I \cdot \cos(\varphi)$$

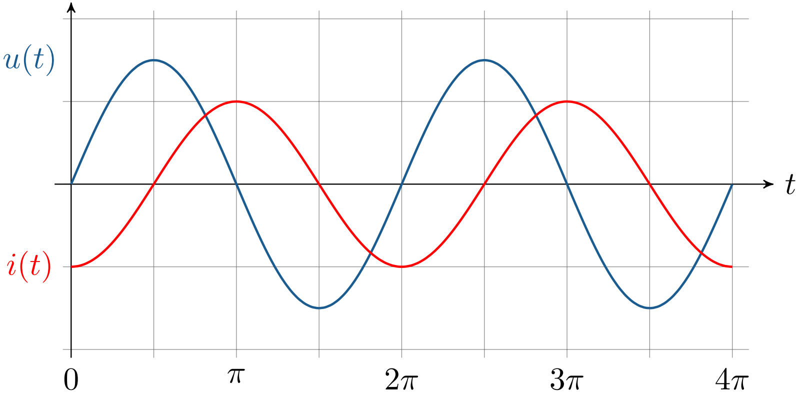
- Average power at an ideal resistance without phase shift between voltage and current:

$$\overline{p}_{\text{Ohm}} = \frac{\hat{U}}{\sqrt{2}} \cdot \frac{\hat{I}}{\sqrt{2}} = \frac{\hat{U} \cdot \hat{I}}{2} = U \cdot I$$

Power Inductance

Instantaneous power at an inductance:

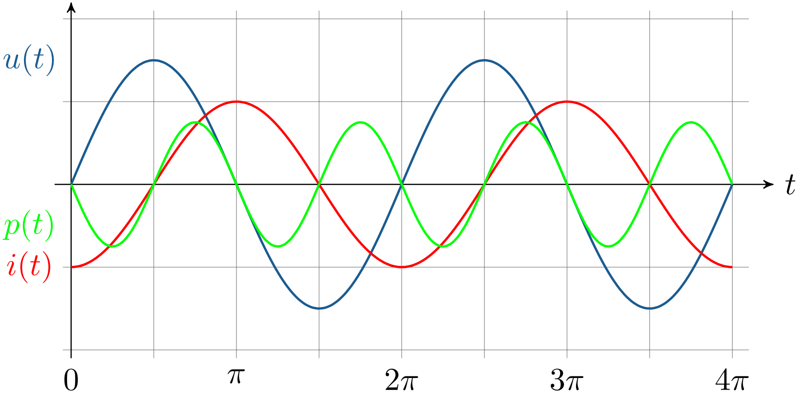
- Voltage and current phase-shifted by $+\pi/2$
- DC component equal to zero



Power Inductance

Instantaneous power at an inductance:

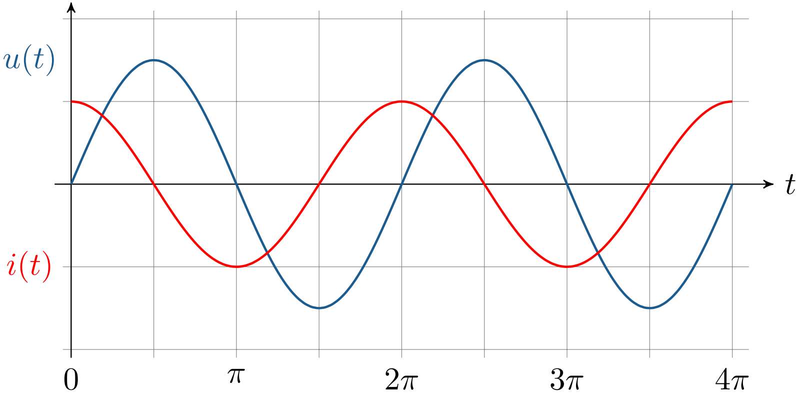
- Voltage and current phase-shifted by $+\pi/2$
- DC component equal to zero
- Instantaneous power pulsates at double frequency
- The power absorbed within one period is completely re-emitted



Power Capacity

Instantaneous power at a given capacity:

- Voltage and current phase-shifted by $-\pi/2$
- Constant portion of instantaneous power equal to zero



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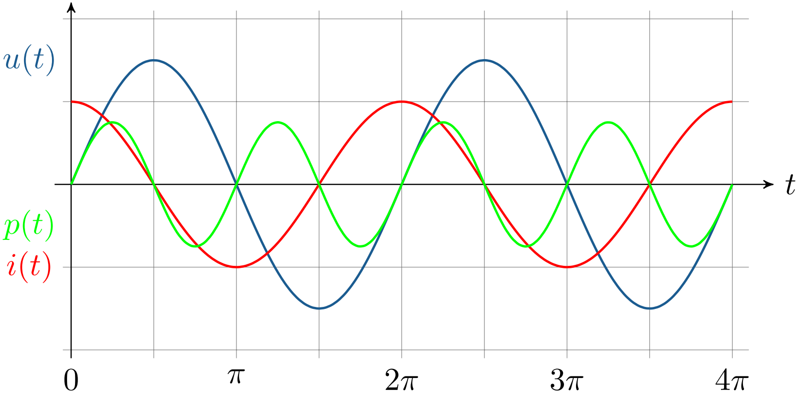

Reactive power

Reactive power describes the portion of power that is required to generate electric and magnetic fields.

- Periodic component of instantaneous power with zero DC component

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$$Q_{\mathrm{Kap}} = U \cdot I \cos(\varphi - 90^\circ) = -U \cdot I \cdot \sin(\varphi)$$

$$[Q] = 1\ Volt - Ampere - reactive = 1\ \mathrm{var}$$

Active power and reactive power

Key point: Active power and reactive power

At the complex level, power is explained by active power and reactive power. Active power describes the real part and reactive power the imaginary part of the complex apparent power. Active power and reactive power are defined as follows:

$$P = U \cdot I \cdot \cos(\varphi) \qquad Q = U \cdot I \cdot \sin(\varphi)$$

Apparent power

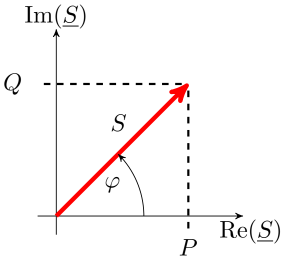
- Apparent power at an ideal resistance:

$$S = U \cdot I = P_{\mathrm{R}}$$

- Complex apparent power:

$$\begin{aligned} \underline{S} &= \underline{U} \cdot \underline{I}^* \\ mit \quad \underline{U} &= U \cdot e^{j\varphi_{\mathrm{u}}} \\ und \quad \underline{I}^* &= I \cdot e^{-j\varphi_{\mathrm{i}}} \\ folgt \quad \underline{S} &= U \cdot e^{j\varphi_{\mathrm{u}}} \cdot I \cdot e^{-j\varphi_{\mathrm{i}}} \\ &= U \cdot I \cdot e^{j(\varphi_{\mathrm{u}} - \varphi_{\mathrm{i}})} \\ &= U \cdot I \cdot e^{j(\varphi)} \end{aligned}$$

- Pointer diagram:



Power types

- From voltage and current to complex apparent power:

$$U \cdot I \cdot e^{j\varphi} = U \cdot I \cdot (\cos\varphi + j \ \sin\varphi) = P + jQ = \underline{S}$$

- Conversion of active power/reactive power and apparent power via power factor:

$$P = \cos(\varphi) \cdot S \qquad\qquad Q = S \cdot \sin(\varphi)$$

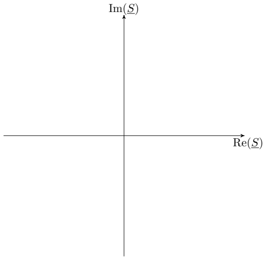
Key point: Komplexe Scheinleistung

The complex apparent power is composed of active power and reactive power. It can be determined in various ways:

$$\underline{S} = \underline{U} \cdot \underline{I}^* = U \cdot I \cdot e^{j(\varphi_\mathrm{u} - \varphi_\mathrm{i})} = P + jQ$$

Operating diagram

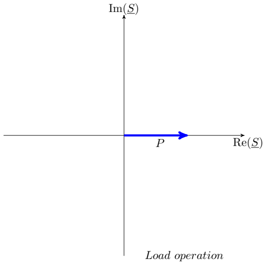
Operating diagram for the complex apparent power of a complex two-pole:



Operating diagram

Operating diagram for the complex apparent power of a complex two-pole:

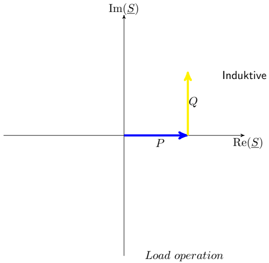
- Positive active power $\rightarrow$ Load operation



Operating diagram

Operating diagram for the complex apparent power of a complex two-pole:

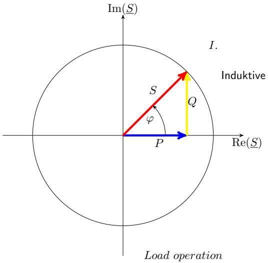
- Positive active power $\rightarrow$ Load operation
- Positive reactive power $\rightarrow$ Inductive



Operating diagram

Operating diagram for the complex apparent power of a complex two-pole:

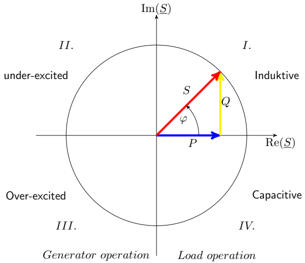
- Positive active power $\rightarrow$ Load operation
- Positive reactive power $\rightarrow$ Inductive
- Apparent power in the first quadrant



Operating diagram

Operating diagram for the complex apparent power of a complex two-pole:

- Positive active power $\rightarrow$ Load operation
- Positive reactive power $\rightarrow$ Inductive
- Apparent power in the first quadrant
- in the second quadrant: Inductive generator operation
- in the third quadrant: Capacitive generator operation
- in the fourth quadrant: Capacitive load operation



Reactive power in generator operation:

- Excitation current of the generator ¿ requirement of the magnetic field $\rightarrow$ overexcited
  Reactive power is fed into the grid $\rightarrow$ voltage increase
  Application: voltage maintenance
- Excitation current of the generator ¡ requirement of the magnetic field $\rightarrow$ underexcited
  Reactive power must be drawn from the grid $\rightarrow$ Voltage reduction
  Possible failure of electrical machines

Summary

Power types at variable values:

- Active power $P = U \cdot I \cdot \cos(\varphi)$: ohmic component
- Reactive power $Q = U \cdot I \cdot \sin(\varphi)$: inductive/capacitive component
- Apparent power $\underline{S} = \underline{U} \cdot \underline{I}^* = P + jQ$: Overall power of the system

Power type		**Unit**	
Active power	P	W	(Watt)
Reactive power	Q	var	(Volt-Ampere-reactiv)
Apparent power	S	VA	(Volt-Ampere)

Power and electrical energy

- Electrical energy:

$$E_{\mathrm{el}} = \int\limits_{t_1}^{t_2} p_{\mathrm{el}}(t)\ \mathrm{d}t$$

- Division into active work and reactive work:

$$W_{\mathrm{N}} = \int\limits_{t_1}^{t_2} p(t)\ \mathrm{d}t \qquad W_{\mathrm{Q}} = \int\limits_{t_1}^{t_2} Q\ \mathrm{d}t$$

- Calculation using effective values:

$$W_{\mathrm{N}} = P \cdot (t_2 - t_1) \qquad W_{\mathrm{Q}} = Q \cdot (t_2 - t_1)$$

Example task

Power calculation:

Given is:

$$\hat{U} = 12\ V \qquad \hat{I} = 2A \qquad \varphi = 60^\circ$$

The following tasks are to be completed:

a) Calculation of active power and reactive power.

b) Calculation of apparent power using the conjugate complex current.

Example task

Power calculation:

a) The active power is:

$$P = U \cdot I \cdot \cos(\varphi) = \frac{\hat{U}}{\sqrt{2}} \cdot \frac{\hat{I}}{\sqrt{2}} \cdot \cos(\varphi) = \frac{12\ V}{\sqrt{2}} \cdot \frac{2\ A}{\sqrt{2}} \cdot \cos(60^o)$$

$$P = 6\ W$$

The reactive power is:

$$Q = U \cdot I \cdot \sin(\varphi) = \frac{\hat{U}}{\sqrt{2}} \cdot \frac{\hat{I}}{\sqrt{2}} \cdot \sin(\varphi) = \frac{12\ V}{\sqrt{2}} \cdot \frac{2\ A}{\sqrt{2}} \cdot \sin(60^o)$$

$$Q = 10{,}392\ var$$

Power calculation:

b) The apparent power is:

$$\underline{S} = \underline{U} \cdot \underline{I}^* = \frac{\underline{\hat{U}}}{\sqrt{2}} \cdot \frac{\underline{\hat{I}}^*}{\sqrt{2}} = \frac{12\ V}{\sqrt{2}} \cdot e^{j(2\pi \cdot 20\ Hz)} \cdot \frac{2\ A}{\sqrt{2}} \cdot e^{j(2\pi \cdot 20\ Hz + \frac{\pi}{3})}$$

$$\underline{S} = 12\ VA \cdot e^{j\frac{\pi}{3}} = 6\ W + j10{,}392\ var$$

Learning objectives: Drehstrom

The students

- understand the basic characteristics of three-phase current and the associated operators.
- can analyse three-phase systems with the same load (symmetrical components).
- are familiar with the characteristics of three-phase systems with different phase loads (unbalanced components).

Symmetrical components

- Phases are generated depending on the number of windings
- In the symmetrical case, the windings are installed in the generator with the same angle α

$$\alpha = \frac{2\pi}{m}$$

- with $m =$ number of phases
- For a three-phase system: $m = 3$

$$\alpha = \frac{2\pi}{3} = 120^\circ$$

The rotation operator a as a function of $\underline{\alpha}$:

$$\underline{a} = e^{j\alpha} = e^{j\frac{2\pi}{m}}$$

For $m = 3$:

$$\begin{aligned}
\underline{a} &= e^{j\frac{2\pi}{3}} = -\frac{1}{2} + j\frac{\sqrt{3}}{2} \\
\underline{a}^2 &= e^{j\frac{4\pi}{3}} = -\frac{1}{2} - j\frac{\sqrt{3}}{2} = \underline{a}^* \\
\underline{a}^3 &= e^{j2\pi} = 1 \\
\underline{a}^4 &= e^{j\frac{8\pi}{3}} = -\frac{1}{2} + j\frac{\sqrt{3}}{2} = \underline{a} \\
\underline{a}^5 &= e^{j\frac{10\pi}{3}} = -\frac{1}{2} - j\frac{\sqrt{3}}{2} = \underline{a}^2 \\
&\vdots
\end{aligned}$$

Key point: Rotation operator

The angle α indicates the phase shift between the phases. With the help of the rotation operator $\underline{a}$, voltages and currents can be converted.

Symmetrical components

The following applies to the voltage of the respective outer conductors relative to a neutral conductor:

$$\underline{U}_{\mathrm{Lm}} = U_{\mathrm{L1}} \cdot e^{-j(m-1)\alpha}$$

Voltage U_m as a function of the rotation operator a:

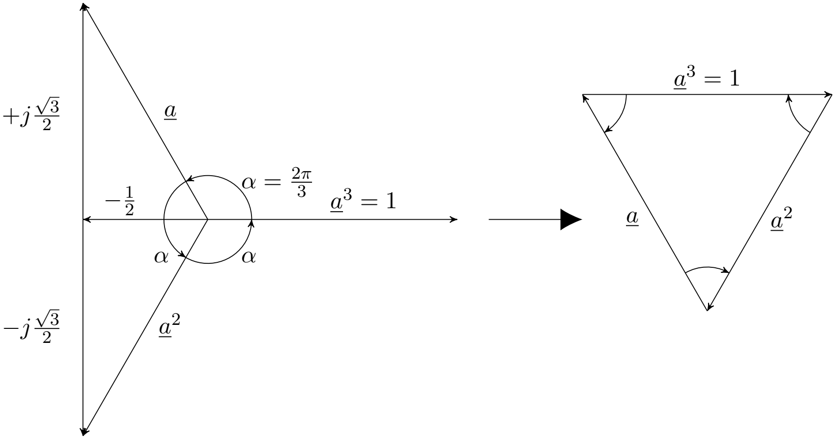
$$\begin{aligned}
\underline{U}_{\mathrm{L1}} &= \underline{a}^{m} \cdot U_{\mathrm{L1}} = 1 \cdot U_{\mathrm{L1}} \\
\underline{U}_{\mathrm{L2}} &= \underline{a}^{m-1} \cdot U_{\mathrm{L1}} \\
&\vdots \\
\underline{U}_{\mathrm{Lm-1}} &= \underline{a}^{2} \cdot U_{\mathrm{L1}} \\
\underline{U}_{\mathrm{Lm}} &= \underline{a} \cdot U_{\mathrm{L1}}
\end{aligned}$$

Symmetrical components

With three phases, i.e. $m = 3$, the following relationship between the three voltages and the rotation operator results:

$$\underline{U}_{\mathrm{L1}} + \underline{U}_{\mathrm{L2}} + \underline{U}_{\mathrm{L3}} = U_1(1 + \underline{a}^2 + \underline{a})$$

Symmetrical components



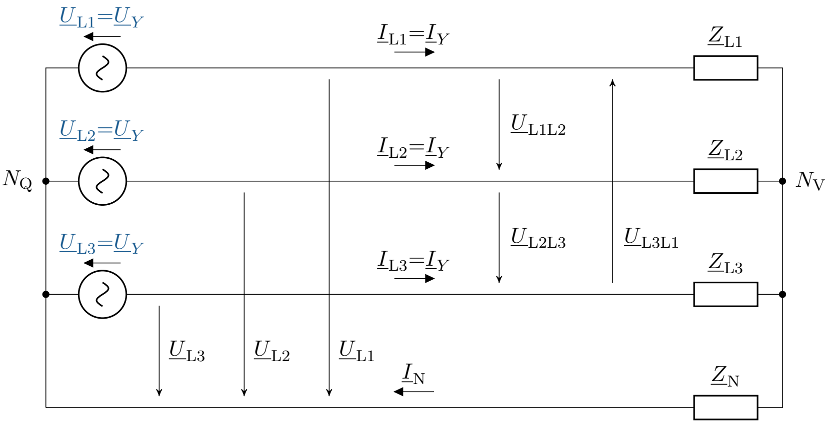
Pointer diagram for the rotation operator → The sum is zero.

$$1 + \underline{a} + \underline{a}^2 = 0$$
$$\underline{U}_{\mathrm{L1}} + \underline{U}_{\mathrm{L2}} + \underline{U}_{\mathrm{L3}} = 0$$

Line quantities in a star (wye) connection

- The generator and consumer sides can be connected either in a star or a delta configuration
- In a star connection, the phases are connected to a star point
- There are then nodes on the generator and consumer sides that can be connected via a neutral conductor
- A voltage source is entered for each phase generated by the generator
- The consumers connected to each phase are combined into one impedance.

Interlinked systems in a star configuration



ESB for a three-phase system with generators and consumers in a star configuration and return conductor

Interlinked systems in a star configuration

Voltages

In principle, there are three voltages in a three-phase system that should be specifically mentioned:

- Phase-to-neutral voltage (star voltage) U_Y: Voltage of the conductor relative to the neutral conductor (U_{L1}, U_{L2}, U_{L3})
- External conductor voltage (chained voltage) U: Voltage between two conductors (U_{L1L2}, U_{L2L3}, U_{L3L1}) – this voltage is also referred to as delta voltage
- Phase voltage U_{str}: Voltage across the impedance of the conductor relative to the neutral conductor. Depending on the connection, the phase voltage is either equal to the star voltage or the delta voltage.

Currents

The current in the neutral conductor is determined by Kirchhoff's 1st law:

$$\underline{I}_{\mathrm{N}} = \underline{I}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}}$$

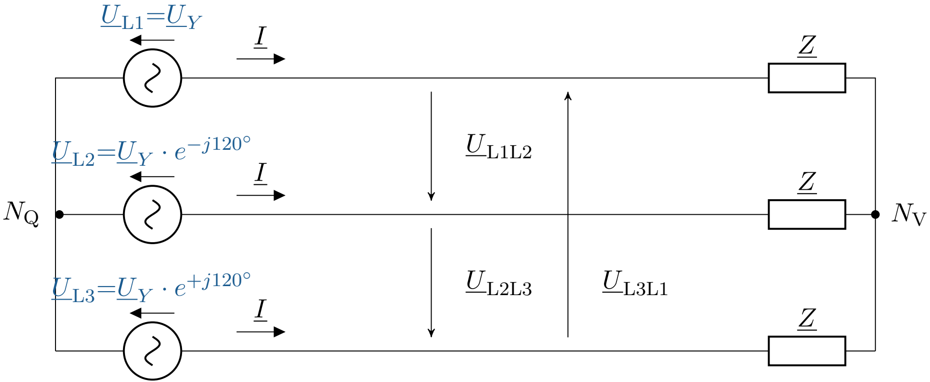
Converted according to Ohm's law:

$$\underline{I}_{\mathrm{N}} = \frac{\underline{U}_{\mathrm{L1}}}{\underline{Z}_1} + \frac{\underline{U}_{\mathrm{L2}}}{\underline{Z}_2} + \frac{\underline{U}_{\mathrm{L3}}}{\underline{Z}_3} = \underline{U}_{\mathrm{1m}}\underline{Y}_1 + \underline{U}_{\mathrm{2m}}\underline{Y}_2 + \underline{U}_{\mathrm{3m}}\underline{Y}_3$$

In the symmetrical case, all impedances are equal ($\underline{Z}_{\mathrm{L1}} = \underline{Z}_{\mathrm{L2}} = \underline{Z}_{\mathrm{L3}} = \underline{Z}$), from which it follows that:

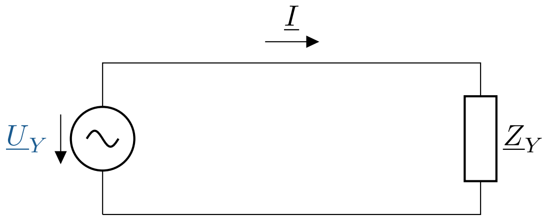
$$\underline{I}_{\mathrm{N}} = \frac{1}{\underline{Z}}(\underline{U}_{\mathrm{L1}} + \underline{U}_{\mathrm{L2}} + \underline{U}_{\mathrm{L3}}) = 0$$

Interlinked systems in a star configuration



Three-phase ESB with symmetrical consumers in star configuration

Interlinked systems in a star configuration



- Alternatively, the ESB can also be set up as single-phase
- It is important to correctly label the currents, voltages, and impedances

Symmetrical components

Key point: Symmetrical components

Symmetrical components describe an **equal load on the phases**, for example in a three-phase system. Here, all three phases are loaded by identical consumers.

Linked systems in a triangle

- In a delta connection, the phases are connected directly to each other
- However, in most cases, the generator side remains connected in a star configuration in order to better dissipate compensation currents
- The outer conductor voltage (also known as delta voltage) is calculated from the difference between the respective star voltages:

$$\underline{U}_{\mathrm{L1L2}} = \underline{U}_{\mathrm{L1}} - \underline{U}_{\mathrm{L2}}$$

Interlinked systems in a triangle

- The relationship between triangular and star tension can be described using a trigonometric derivation:

$$\sin\left(\frac{\alpha}{2}\right) = \sin\left(\frac{\frac{2\pi}{m}}{2}\right) = \sin\left(\frac{\pi}{m}\right) = \frac{G}{H}$$

$$\sin\left(\frac{\pi}{3}\right) = \frac{\frac{\underline{U}_{\mathrm{L1L2}}}{2}}{\underline{U}_{\mathrm{L1}}}$$

$$2 \cdot \frac{\sqrt{3}}{2} = \frac{\underline{U}_{\mathrm{L1L2}}}{\underline{U}_{\mathrm{L1}}}$$

$$\underline{U}_{\mathrm{L1}} = \frac{\underline{U}_{\mathrm{L1L2}}}{\sqrt{3}}$$

Star-delta conversion factor

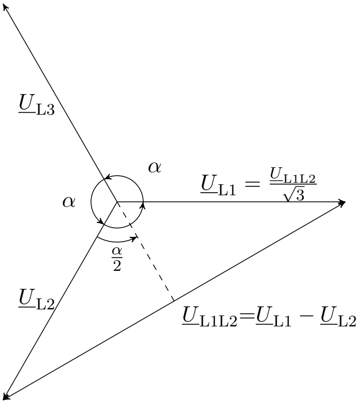
Key point: Stern-Dreieck-Umrechnungsfaktor

The star voltage and the delta voltage can be converted into each other using the conversion factor $\sqrt{3}$:

$$U_Y = \frac{U_\Delta}{\sqrt{3}}$$

Interlinked systems in a triangle

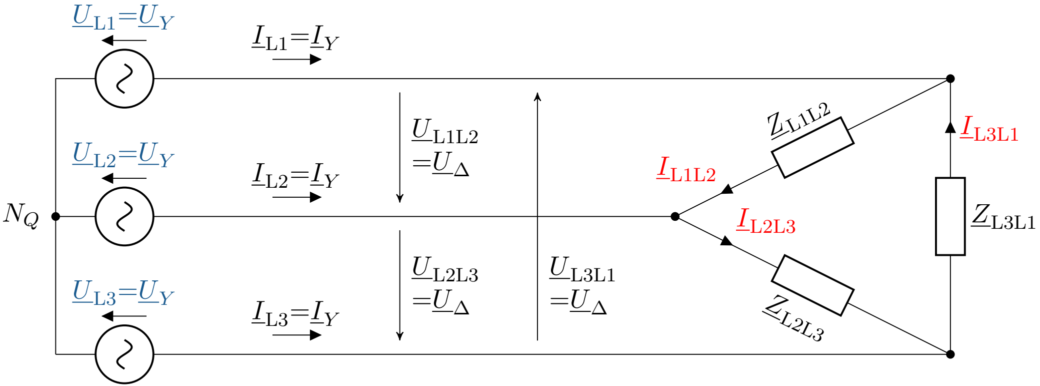
- The derivation can be illustrated even better on the basis of the pointer diagram:



- In practice, the triangle symbol is usually omitted from the index:

$$U_Y = \frac{U_\Delta}{\sqrt{3}} = \frac{U}{\sqrt{3}}$$

Three-phase ESB with generators in star configuration and consumers in delta configuration



Interlinked systems in a triangle

- In a delta connection, the phase voltage corresponds to the outer conductor (delta) voltage
- In the symmetrical case, the impedances are also equal in this circuit
- The following equations can be established for the current:

$$\underline{I}_{\mathrm{L1}} = \underline{I}_{\mathrm{L1L2}} - \underline{I}_{\mathrm{L3L1}}$$
$$\underline{I}_{\mathrm{L2}} = \underline{I}_{\mathrm{L2L3}} - \underline{I}_{\mathrm{L1L2}}$$
$$\underline{I}_{\mathrm{L3}} = \underline{I}_{\mathrm{L3L1}} - \underline{I}_{\mathrm{L2L3}}$$

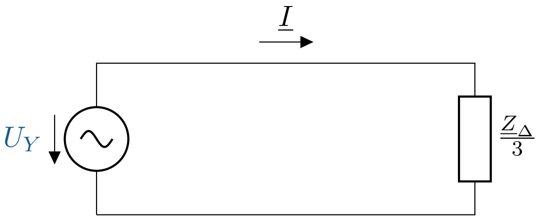
$$\underline{I}_{\mathrm{L1}} = \underline{I}_{\mathrm{L1L2}} - \underline{I}_{\mathrm{L3L1}} = \frac{\underline{U}_{\mathrm{L1L2}} - \underline{U}_{\mathrm{L3L1}}}{\underline{Z}} = \frac{\underline{U}_{Y}}{\underline{Z}} \cdot (1 - \underline{a}^2 - \underline{a} + 1) = 3 \cdot \frac{\underline{U}_{Y}}{\underline{Z}}$$
$$\underline{I}_{\mathrm{L2}} = \underline{I}_{\mathrm{L2L3}} - \underline{I}_{\mathrm{L1L2}} = \frac{\underline{U}_{\mathrm{L2L3}} - \underline{U}_{\mathrm{L1L2}}}{\underline{Z}} = \frac{\underline{U}_{Y}}{\underline{Z}} \cdot (\underline{a}^2 - \underline{a} - 1 + \underline{a}^2) = 3 \cdot \underline{a}^2 \cdot \frac{\underline{U}_{Y}}{\underline{Z}}$$
$$\underline{I}_{\mathrm{L3}} = \underline{I}_{\mathrm{L3L1}} - \underline{I}_{\mathrm{L2L3}} = \frac{\underline{U}_{\mathrm{L3L1}} - \underline{U}_{\mathrm{L2L3}}}{\underline{Z}} = \frac{\underline{U}_{Y}}{\underline{Z}} \cdot (\underline{a} - 1 - \underline{a}^2 + \underline{a}) = 3 \cdot \underline{a} \cdot \frac{\underline{U}_{Y}}{\underline{Z}}$$

Interlinked systems in a triangle

- The ratio of phase current to delta current can be calculated in the same way as the ratio of star voltage to delta voltage:

$$I_{\text{str}} = \sqrt{3} \cdot I_{\Delta}$$

- A single-phase ESB can also be set up for the delta connection:



Power in a symmetrical three-phase network

- Power is calculated using the equation $P = U \cdot I$
- In three-phase current, the power must be calculated for each phase
- In a symmetrical case, the currents and voltage in the phases are equal, so the following applies:

$$
\begin{aligned}
S &= 3 \cdot I_{\text{str}} \cdot U_Y = 3 \cdot I_{\text{str}} \cdot \frac{U_\Delta}{\sqrt{3}} \\
&= 3 \cdot I_\Delta \cdot U_\Delta = 3 \cdot \frac{I_{\text{str}}}{\sqrt{3}} \cdot U_\Delta \\
&= \sqrt{3} \cdot U_\Delta \cdot I_{\text{str}}
\end{aligned}
$$

- Fundamentally, it makes no difference which circuit is used for the calculation; only the appropriate values need to be used.

Power in a symmetrical three-phase AC network

- With the same impedance, the power in the delta connection is three times greater than in the star connection:

Star connection:

$$S = \sqrt{3} \cdot U \cdot I = \sqrt{3} \cdot U \cdot \frac{U_Y}{Z} = \frac{U^2}{Z}$$

Triangular connection:

$$S = \sqrt{3} \cdot U \cdot I = \sqrt{3} \cdot U \cdot \sqrt{3} \cdot \frac{U}{Z} = 3 \cdot \frac{U^2}{Z}$$

Unbalanced components

- Until now, it has been assumed that generators and consumers are symmetrical
- Now let us consider how the system behaves when asymmetries are present
- However, the generator side is still assumed to be symmetrical and connected in a star configuration
- For consumers, a distinction is made between four-wire networks in star configuration, three-wire networks in star configuration, and three-wire networks in delta configuration

Asymmetrical components

Key point: Unbalanced components

Unbalanced components unlike symmetrical components, they describe the **uneven load on the phases** due to different consumers.

Four-wire network in star connection

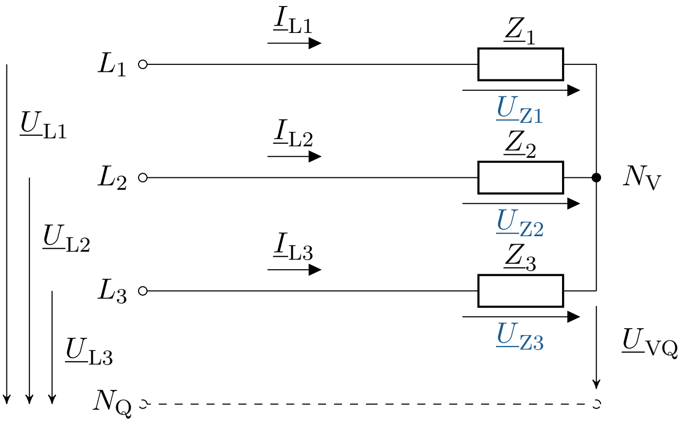
- The structure is the same as for the ESB from the chapter on star-connected systems
- The return conductor connects the star points on the generator and consumer sides
- If the generator side remains symmetrical, the star voltages remain symmetrical
- The currents adjust according to the different impedances
- The following applies to the current in the return conductor:

$$\underline{I}_{\mathrm{N}} = \underline{I}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}}$$

- The power is calculated using the familiar formula
- In the unbalanced case, it must be calculated individually for each conductor.

$$\underline{S} = \underline{I}^{*}_{\mathrm{L1}} \cdot \underline{U}_{\mathrm{L1}} + \underline{I}^{*}_{\mathrm{L2}} \cdot \underline{U}_{\mathrm{L2}} + \underline{I}^{*}_{\mathrm{L3}} \cdot \underline{U}_{\mathrm{L3}}$$

Simplified ESB: Three-wire network in star connection with unbalanced consumers



Three-wire network in star connection

- Similar conditions as for the four-wire network in star connection, but without a return conductor
- This prevents the current from flowing back via the return conductor and therefore affects the other phases
- Accordingly, the voltage drop between the star points and across the loads must also be calculated
- The values can be determined using two methods:

1. Convert impedances mathematically into a delta connection and calculate them using the known methods

- Because: In a delta connection, the voltages across the impedances are equal to the conductor voltages

2. Set up equations using Ohm's and Kirchhoff's laws

- The following equation can be set up for the difference voltage between the two star points:

$$\underline{U}_{VQ} = \frac{\frac{\underline{U}_{\mathrm{L1}}}{\underline{Z}_1} + \frac{\underline{U}_{\mathrm{L2}}}{\underline{Z}_2} + \frac{\underline{U}_{\mathrm{L3}}}{\underline{Z}_3}}{\frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2} + \frac{1}{\underline{Z}_3}} = \frac{\underline{U}_{\mathrm{L1}} \cdot \underline{Z}_2 \cdot \underline{Z}_3 + \underline{U}_{\mathrm{L2}} \cdot \underline{Z}_3 \cdot \underline{Z}_1 + \underline{U}_{\mathrm{L3}} \cdot \underline{Z}_1 \cdot \underline{Z}_2}{\underline{Z}_1 \cdot \underline{Z}_2 + \underline{Z}_2 \cdot \underline{Z}_3 + \underline{Z}_3 \cdot \underline{Z}_1}$$

- The power is calculated in each phase; it is important to measure the voltage across the consumers.

$$\underline{S} = \underline{I}^*_{\mathrm{L1}} \cdot \underline{U}_{\mathrm{Z1}} + \underline{I}^*_{\mathrm{L2}} \cdot \underline{U}_{\mathrm{Z2}} + \underline{I}^*_{\mathrm{L3}} \cdot \underline{U}_{\mathrm{Z3}}$$

- The equation for calculating power can be rearranged and simplified

$$\begin{aligned}\underline{S} &= \underline{I}^*_{\mathrm{L1}} \cdot \underline{U}_{\mathrm{L1}} + \underline{I}^*_{\mathrm{L2}} \cdot \underline{U}_{\mathrm{L2}} + \underline{I}^*_{\mathrm{L3}} \cdot \underline{U}_{\mathrm{L3}} - \underline{U}_{\mathrm{VQ}} \cdot (\underline{I}^*_{\mathrm{L1}} + \underline{I}^*_{\mathrm{L2}} + \underline{I}^*_{\mathrm{L3}}) \\ &= \underline{I}^*_{\mathrm{L1}} \cdot \underline{U}_{\mathrm{L1}} + \underline{I}^*_{\mathrm{L2}} \cdot \underline{U}_{\mathrm{L2}} + \underline{I}^*_{\mathrm{L3}} \cdot \underline{U}_{\mathrm{L3}}\end{aligned}$$

- Using the node rule, one of the currents can in turn be expressed by the other two

$$\begin{aligned}\underline{S} &= \underline{I}^*_{\mathrm{L1}} \cdot \underline{U}_{\mathrm{L1}} + \underline{I}^*_{\mathrm{L2}} \cdot \underline{U}_{\mathrm{L2}} + (-\underline{I}^*_{\mathrm{L1}} - \underline{I}^*_{\mathrm{L2}}) \cdot \underline{U}_{\mathrm{L3}} \\ &= (\underline{U}_{\mathrm{L1}} - \underline{U}_{\mathrm{L2}}) \cdot \underline{I}^*_{\mathrm{L1}} + (\underline{U}_{\mathrm{L2}} - \underline{U}_{\mathrm{L3}}) \cdot \underline{I}^*_{\mathrm{L2}} \\ &= \underline{U}_{\mathrm{L3L1}} \cdot \underline{I}^*_{\mathrm{L1}} + \underline{U}_{\mathrm{L2L3}} \cdot \underline{I}^*_{\mathrm{L2}}\end{aligned}$$

Three-wire network in delta connection

- There is no neutral point available in the delta connection, so it can only be implemented in a three-wire system.
- Due to the symmetrical feed, the outer conductor voltages are symmetrical.
- The currents adjust to the respective impedance:

$$\underline{I}_{\mathrm{L1}} = \underline{I}_{\mathrm{L1L2}} - \underline{I}_{\mathrm{L3L1}}$$
$$\underline{I}_{\mathrm{L2}} = \underline{I}_{\mathrm{L2L3}} - \underline{I}_{\mathrm{L1L2}}$$
$$\underline{I}_{\mathrm{L3}} = \underline{I}_{\mathrm{L3L1}} - \underline{I}_{\mathrm{L2L3}}$$

Three-wire network in delta connection

- The sum of the external conductor currents must always be zero, even with unbalanced currents.
- This results in the same power calculation as for the star connection in a three-wire network and therefore also the same transformations.

$$\begin{aligned} \underline{S} &= \underline{I}^*_{\mathrm{L1L2}} \cdot \underline{U}_{\mathrm{L1L2}} + \underline{I}^*_{\mathrm{L2L3}} \cdot \underline{U}_{\mathrm{L2L3}} + \underline{I}^*_{\mathrm{L3L1}} \cdot \underline{U}_{\mathrm{L3L1}} \\ &= \underline{I}^*_{\mathrm{L1}} \cdot \underline{U}_{\mathrm{L1}} + \underline{I}^*_{\mathrm{L2}} \cdot \underline{U}_{\mathrm{L2}} + \underline{I}^*_{\mathrm{L3}} \cdot \underline{U}_{\mathrm{L3}} \end{aligned}$$

$$\begin{aligned} \underline{S} &= \underline{I}^*_{\mathrm{L1}} \cdot \underline{U}_{\mathrm{L1}} + \underline{I}^*_{\mathrm{L2}} \cdot \underline{U}_{\mathrm{L2}} + (-\underline{I}^*_{\mathrm{L1}} - \underline{I}^*_{\mathrm{L2}}) \cdot \underline{U}_{L3} \\ &= (\underline{U}_{\mathrm{L1}} - \underline{U}_{\mathrm{L2}}) \cdot \underline{I}^*_{\mathrm{L1}} + (\underline{U}_{\mathrm{L2}} - \underline{U}_{\mathrm{L3}}) \cdot \underline{I}^*_{\mathrm{L2}} \\ &= \underline{U}_{\mathrm{L3L1}} \cdot \underline{I}^*_{\mathrm{L1}} + \underline{U}_{\mathrm{L2L3}} \cdot \underline{I}^*_{\mathrm{L2}} \end{aligned}$$

Learning objectives: Multiphase systems

The students

- understand the functions of shunt, counter and zero systems.
- can calculate the three-phase power of the various systems.
- can create equivalent circuit diagrams for symmetrical sources, loads and lines.

In order to make meaningful use of such a transformation, certain requirements must be met:

1. Calculations in the transformed system must be simpler than in the original system
2. The symmetries of the real networks and equipment must be used to advantage
3. Both currents and voltages must be treated with the same transformation rules.
4. The power ratings should be identical in both systems; if this is not possible, the power ratings should be convertible by a constant factor.

Positive-, negative-, and zero-sequence systems

- The aim of the transformation is to convert an asymmetrical system of n vectors into n systems with a symmetrical vector arrangement
- In three-phase current, each conductor (L_1, L_2, L_3) is assigned a system with a symmetrical vector arrangement
- With the help of the rotation operator, symmetry conditions can be established for each conductor, known as decomposition equations:

$$\underline{I}_{\mathrm{L1}} = \underline{I}_0 + \underline{I}_1 + \underline{I}_2$$
$$\underline{I}_{\mathrm{L2}} = \underline{I}_0 + \underline{a}^2 \cdot \underline{I}_1 + \underline{a} \cdot \underline{I}_2$$
$$\underline{I}_{\mathrm{L3}} = \underline{I}_0 + \underline{a} \cdot \underline{I}_1 + \underline{a}^2 \cdot \underline{I}_2$$

It should be noted here that everything that applies to current also applies to voltage in the same context!

- In the next step, the decomposition equations should be equated and converted to the current of the respective system
- For the zero system, this results in the following equation:

$$
\begin{aligned}
\underline{I}_{\mathrm{L}1} + \underline{I}_{\mathrm{L}2} + \underline{I}_{\mathrm{L}3} &= 3 \cdot \underline{I}_0 + \underline{I}_1 \cdot (1 + \underline{a}^2 + \underline{a}) + \underline{I}_2 \cdot (1 + \underline{a} + \underline{a}^2) \\
&= 3 \cdot \underline{I}_0 \\
\Leftrightarrow \underline{I}_0 &= \frac{1}{3} \cdot (\underline{I}_{\mathrm{L}1} + \underline{I}_{\mathrm{L}2} + \underline{I}_{\mathrm{L}3})
\end{aligned}
$$

- To convert the added currents for the current of the secondary system, an intermediate step must be taken
- First, the rotation operator must be calculated from the prefactor by multiplying by the appropriate rotation operators:

$$
\begin{aligned}
\underline{I}_{\mathrm{L1}} &= \underline{I}_0 + \underline{I}_1 + \underline{I}_2 \\
\underline{a} \cdot \underline{I}_{\mathrm{L2}} &= \underline{a} \cdot \underline{I}_0 + \underline{a}^3 \cdot \underline{I}_1 + \underline{a}^2 \cdot \underline{I}_2 \\
\underline{a}^2 \cdot \underline{I}_{\mathrm{L3}} &= \underline{a}^2 \cdot \underline{I}_0 + \underline{a}^3 \cdot \underline{I}_1 + \underline{a} \cdot \underline{I}_2
\end{aligned}
$$

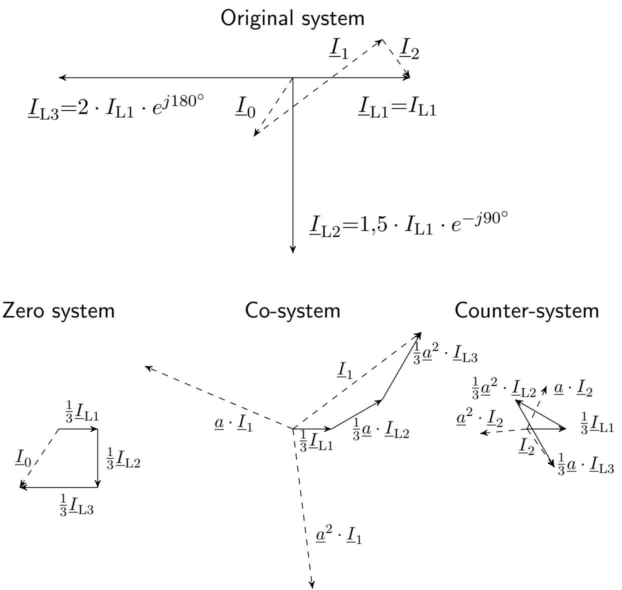
$$
\begin{aligned}
\underline{I}_{\mathrm{L1}} + \underline{a} \cdot \underline{I}_{\mathrm{L2}} + \underline{a}^2 \cdot \underline{I}_{\mathrm{L3}} &= \underline{I}_0 \cdot (1 + \underline{a} + \underline{a}^2) + 3 \cdot \underline{I}_1 + \underline{I}_2 \cdot (1 + \underline{a}^2 + \underline{a}) \\
&= 3 \cdot \underline{I}_1 \\
\Leftrightarrow \underline{I}_1 &= \frac{1}{3} \cdot (\underline{I}_{L1} + \underline{a} \cdot \underline{I}_{L2} + \underline{a}^2 \cdot \underline{I}_{L3})
\end{aligned}
$$

- The current for the opposite system is calculated in the same way as for the same system:

$$\begin{aligned}
\underline{I}_{\mathrm{L}1} &= \underline{I}_0 + \underline{I}_1 + \underline{I}_2 \\
\underline{a}^2 \cdot \underline{I}_{\mathrm{L}2} &= \underline{a}^2 \cdot \underline{I}_0 + \underline{a}^4 \cdot \underline{I}_1 + \underline{a}^3 \cdot \underline{I}_2 \\
\underline{a} \cdot \underline{I}_{\mathrm{L}3} &= \underline{a} \cdot \underline{I}_0 + \underline{a}^2 \cdot \underline{I}_1 + \underline{a}^3 \cdot \underline{I}_2
\end{aligned}$$

$$\begin{aligned}
\underline{I}_{\mathrm{L}1} + \underline{a}^2 \cdot \underline{I}_{\mathrm{L}2} + \underline{a} \cdot \underline{I}_{\mathrm{L}3} &= (1 + \underline{a}^2 + \underline{a})\underline{I}_0 + (1 + \underline{a} + \underline{a}^2) \cdot \underline{I}_1 + 3 \cdot \underline{I}_2 \\
&= 3 \cdot \underline{I}_2 \\
\Leftrightarrow \underline{I}_2 &= \frac{1}{3} \cdot (\underline{I}_{\mathrm{L}1} + \underline{a}^2 \cdot \underline{I}_{\mathrm{L}2} + \underline{a} \cdot \underline{I}_{\mathrm{L}3})
\end{aligned}$$

Pointer diagram of an unbalanced current in a three-phase system and the transformations



- For simplified notation, the transformation equations can be described in matrix notation:

$$\begin{bmatrix}\underline{I}_0\\\underline{I}_1\\\underline{I}_2\end{bmatrix} = \frac{1}{3}\cdot\begin{bmatrix}1 & 1 & 1\\1 & \underline{a} & \underline{a}^2\\1 & \underline{a}^2 & \underline{a}\end{bmatrix}\cdot\begin{bmatrix}\underline{I}_{\mathrm{L1}}\\\underline{I}_{\mathrm{L2}}\\\underline{I}_{\mathrm{L3}}\end{bmatrix}$$

- The notation can be simplified further by combining the term with the rotation operators and the term with the prefactor into a matrix:

$$[\underline{T}] = \frac{1}{3}\cdot\begin{bmatrix}1 & 1 & 1\\1 & \underline{a} & \underline{a}^2\\1 & \underline{a}^2 & \underline{a}\end{bmatrix}$$

Fundamentals - Matrix notation

- When the equations are combined, the following equation is obtained in compact notation:

$$\left[\underline{I}_{012}\right] = \left[\underline{T}\right] \cdot \left[\underline{I}_{\mathrm{L1L2L3}}\right]$$

- According to the rules for calculating matrices, the inverse of $\left[\underline{T}\right]$ can be used to perform the inverse transformation:

$$\begin{bmatrix} \underline{I}_{\mathrm{L1}} \\ \underline{I}_{\mathrm{L2}} \\ \underline{I}_{\mathrm{L3}} \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & \underline{a} & \underline{a}^2 \end{bmatrix} \cdot \begin{bmatrix} \underline{I}_0 \\ \underline{I}_1 \\ \underline{I}_2 \end{bmatrix}$$

$$\left[\underline{I}_{\mathrm{L1L2L3}}\right] = \left[\underline{T}\right]^{-1} \cdot \left[\underline{I}_{012}\right]$$

Types of systems

Key point: Positive-, negative-, and zero-sequence systems

A system is transformed into three systems for analysis: the partner system, the counter system, and the zero system.
The **Positive system** is assigned the index 1: $\underline{I}_1$
The **Negative system** is assigned the index 2: $\underline{I}_2$
The **Zero system** is assigned the index 0: $\underline{I}_0$

Three-phase power

- The equation for calculating power can also be expressed in matrix notation for simplicity.

$$
\begin{aligned}
\underline{S}_{\mathrm{L1L2L3}} &= \underline{U}_{\mathrm{L1}} \cdot \underline{I}^*_{\mathrm{L1}} + \underline{U}_{\mathrm{L2}} \cdot \underline{I}^*_{\mathrm{L2}} + \underline{U}_{\mathrm{L3}} \cdot \underline{I}^*_{\mathrm{L3}} \\
&= \begin{bmatrix} \underline{U}_{\mathrm{L1}} & \underline{U}_{\mathrm{L2}} & \underline{U}_{\mathrm{L3}} \end{bmatrix} \cdot \begin{bmatrix} \underline{I}^*_{\mathrm{L1}} \\ \underline{I}^*_{\mathrm{L2}} \\ \underline{I}^*_{\mathrm{L3}} \end{bmatrix} = \begin{bmatrix} \underline{U}_{\mathrm{L1L2L3}} \end{bmatrix} \cdot \begin{bmatrix} \underline{I}_{\mathrm{L1L2L3}} \end{bmatrix}^*_t
\end{aligned}
$$

- The rotation operators can be used to relate the voltages and currents to a single phase
- This results in the familiar simplification:

$$\begin{aligned} \underline{S}_{\mathrm{L1L2L3}} &= \underline{U}_{\mathrm{L1}} \cdot \underline{I}_{\mathrm{L1}}^{*} + \underline{U}_{\mathrm{L1}} \cdot \underline{I}_{\mathrm{L1}}^{*} + \underline{U}_{\mathrm{L1}} \cdot \underline{I}_{\mathrm{L1}}^{*} \\ &= 3 \cdot \underline{U}_{\mathrm{L1}} \cdot \underline{I}_{\mathrm{L1}}^{*} \end{aligned}$$

- Now, using matrix notation, the transformation into the new systems can be calculated:

$$
\begin{aligned}
[\underline{U}_{\mathrm{L1L2L3}}] &= ([\underline{T}]^{-1} \cdot [\underline{U}_{012}]) = [\underline{U}_{012}] \cdot [\underline{T}]^{-1} \\
[\underline{I}_{\mathrm{L1L2L3}}]^{*} &= ([\underline{T}]^{-1} \cdot [\underline{I}_{012}])^{*} = [\underline{T}]^{-1*} \cdot [\underline{I}_{012}]^{*} \\
\underline{S}_{\mathrm{L1L2L3}} &= [\underline{U}_{012}]_t \cdot [\underline{T}]^{-1} \cdot [\underline{T}]^{-1*} \cdot [\underline{I}_{012}]^{*} \\
&= [\underline{U}_{012}]_t \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & \underline{a} & \underline{a}^2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & \underline{a} & \underline{a}^2 \end{bmatrix}^{*} \cdot [\underline{I}_{012}]^{*} \\
&= [\underline{U}_{012}]_t \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & \underline{a} & \underline{a}^2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \end{bmatrix} \cdot [\underline{I}_{012}]^{*} \\
&= [\underline{U}_{012}] \cdot \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \cdot [\underline{I}_{012}]^{*} \\
&= 3 \cdot (\underline{U}_0 \cdot \underline{I}_o^{*} + \underline{U}_1 \cdot \underline{I}_1^{*} + \underline{U}_2 \cdot \underline{I}_2^{*}) = 3 \cdot \underline{S}_{012}
\end{aligned}
$$

Equivalent circuit diagram

- ESBs essentially serve to simplify the display and understanding of components.
- We will now look at various operating resources and the behavior of transformations in different network situations.

Symmetrical voltage source

- The voltage sources are considered as the first component
- The relationships between the star voltage and the individual phases are already known
- The formula for the transformation is used and converted into matrix notation

$$\begin{bmatrix} \underline{U}_0 \\ \underline{U}_1 \\ \underline{U}_2 \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \end{bmatrix} \cdot \begin{bmatrix} \underline{U}_{\mathrm{L1}} \\ \underline{U}_{\mathrm{L2}} \\ \underline{U}_{\mathrm{L3}} \end{bmatrix}$$

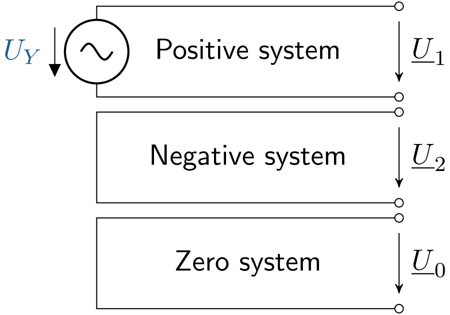
$$[\underline{U}_{012}] = [\underline{T}] \cdot \begin{bmatrix} \underline{U}_Y \\ \underline{a}^2 \cdot \underline{U}_Y \\ \underline{a} \cdot \underline{U}_Y \end{bmatrix}$$

- Solving the equation for the transformed systems yields the voltage results

$$\underline{U}_0 = \frac{1}{3} \cdot U_Y \cdot (1 + \underline{a}^2 + \underline{a}) = 0$$
$$\underline{U}_1 = \frac{1}{3} \cdot U_Y \cdot (1 + \underline{a} \cdot \underline{a}^2 + \underline{a}^2 \cdot \underline{a}) = U_Y$$
$$\underline{U}_2 = \frac{1}{3} \cdot U_Y \cdot (1 + \underline{a}^2 \cdot \underline{a}^2 + \underline{a} \cdot \underline{a}) = 0$$

Symmetrical voltage source

- The ESB of the individual systems can be set up according to the voltage
- With the results of the counter and zero systems, it can be said figuratively that these are short-circuited



Symmetrical load

- For the consideration of the loads, a star connection with a return conductor should be selected
- An impedance Z is assumed in each conductor, including the return conductor
- For the symmetrical case, the impedances are assumed to be equal; only the impedance in the return conductor may differ from the others and is given its own index.

$$\underline{U}_{\mathrm{L1}} = \underline{Z} \cdot \underline{I}_{\mathrm{L1}} + \underline{Z}_{\mathrm{E}} \cdot (\underline{I}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}})$$
$$\underline{U}_{\mathrm{L2}} = \underline{Z} \cdot \underline{I}_{\mathrm{L2}} + \underline{Z}_{\mathrm{E}} \cdot (\underline{I}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}})$$
$$\underline{U}_{\mathrm{L3}} = \underline{Z} \cdot \underline{I}_{\mathrm{L3}} + \underline{Z}_{\mathrm{E}} \cdot (\underline{I}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}})$$

Symmetrical load

- Here, too, matrix notation can be used to simplify the expression.

$$\begin{bmatrix} \underline{U}_{\mathrm{L1}} \\ \underline{U}_{\mathrm{L2}} \\ \underline{U}_{\mathrm{L3}} \end{bmatrix} = \begin{bmatrix} \underline{Z}+\underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} \\ \underline{Z}_{\mathrm{E}} & \underline{Z}+\underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} \\ \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} & \underline{Z}+\underline{Z}_{\mathrm{E}} \end{bmatrix} \cdot \begin{bmatrix} \underline{I}_{\mathrm{L1}} \\ \underline{I}_{\mathrm{L2}} \\ \underline{I}_{\mathrm{L3}} \end{bmatrix}$$

With...

$$\left[\underline{Z}_{\mathrm{L1L2L3}}\right] = \begin{bmatrix} \underline{Z}+\underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} \\ \underline{Z}_{\mathrm{E}} & \underline{Z}+\underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} \\ \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} & \underline{Z}+\underline{Z}_{\mathrm{E}} \end{bmatrix}$$

...results in

$$\left[\underline{U}_{\mathrm{L1L2L3}}\right] = \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \cdot \left[\underline{I}_{\mathrm{L1L2L3}}\right]$$

- If transformed using the transformation equations, the following results:

$$\left[\underline{T}\right]^{-1} \cdot \left[\underline{U}_{012}\right] = \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \left[\underline{T}\right]^{-1} \cdot \left[\underline{I}_{012}\right]$$

- As with voltage sources, the voltage should be calculated from the transformation depending on the original system.
- Conversion creates a relationship between the transformed impedances and the impedances of the original system.

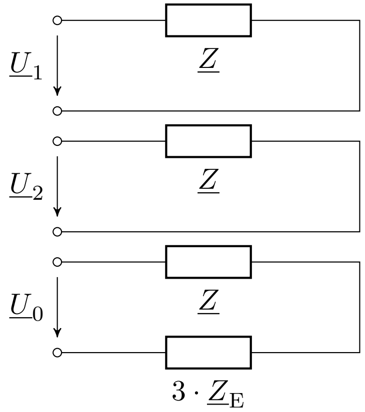
$$\begin{aligned}
\left[\underline{T}\right] \cdot \left[\underline{T}\right]^{-1} \cdot \left[\underline{U}_{012}\right] &= \left[\underline{T}\right] \cdot \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \left[\underline{T}\right]^{-1} \cdot \left[\underline{I}_{012}\right] \\
\left[\underline{U}_{012}\right] &= \left[\underline{T}\right] \cdot \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \left[\underline{T}\right]^{-1} \cdot \left[\underline{I}_{012}\right] \\
\left[\underline{Z}_{012}\right] &= \left[\underline{T}\right] \cdot \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \left[\underline{T}\right]^{-1}
\end{aligned}$$

- Performing the matrix operation yields the desired relationship between the original and transformation systems.

$$\left[\underline{Z}_{012}\right] = \begin{bmatrix} \underline{Z} + 3\underline{Z}_E & 0 & 0 \\ 0 & \underline{Z} & 0 \\ 0 & 0 & \underline{Z} \end{bmatrix}$$

Symmetrical load

- The ESB of the individual systems can be established using the calculated relationships
- The same impedances occur in the positive and negative systems
- In addition, an impedance with triple the value occurs in the zero system



Symmetrical lines

- The line is the component that connects the voltage source and the consumers
- Intrinsic impedances and coupling impedances always occur (conductor-ground impedances are neglected)
- Accordingly, each phase has one intrinsic impedance and two coupling impedances (to the other phases)
- The voltage across a phase is defined as the voltage difference between the voltage at the beginning and end of the line

$$\begin{bmatrix} \underline{U}_{\mathrm{L1A}} - \underline{U}_{\mathrm{L1B}} \\ \underline{U}_{\mathrm{L2A}} - \underline{U}_{\mathrm{L2B}} \\ \underline{U}_{\mathrm{L3A}} - \underline{U}_{\mathrm{L3B}} \end{bmatrix} = \begin{bmatrix} \underline{Z}_{\mathrm{S}} & \underline{Z}_{\mathrm{K}} & \underline{Z}_{\mathrm{K}} \\ \underline{Z}_{\mathrm{K}} & \underline{Z}_{\mathrm{S}} & \underline{Z}_{\mathrm{K}} \\ \underline{Z}_{\mathrm{K}} & \underline{Z}_{\mathrm{K}} & \underline{Z}_{\mathrm{S}} \end{bmatrix} \cdot \begin{bmatrix} \underline{I}_{\mathrm{L1}} \\ \underline{I}_{\mathrm{L2}} \\ \underline{I}_{\mathrm{L3}} \end{bmatrix}$$

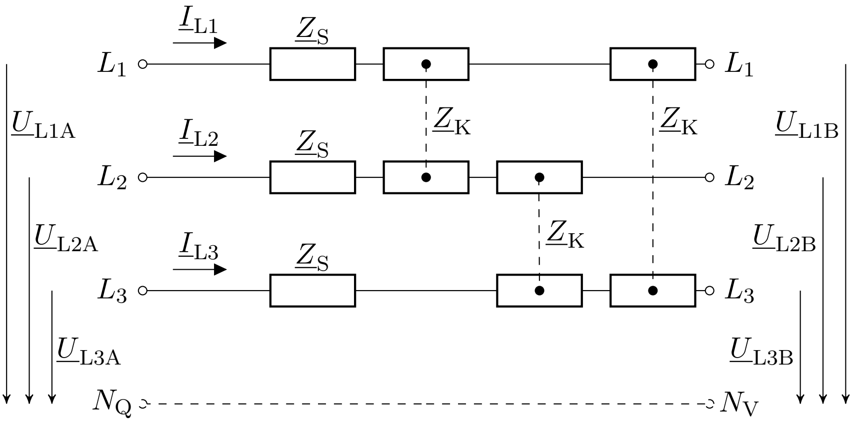
- With $U_{\mathrm{L1A}} - U_{\mathrm{L1B}} = U_{\mathrm{L1}}$, ... a compact notation can be derived again

$$\left[\underline{U}_{\mathrm{L1L2L3}}\right] = \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \cdot \left[\underline{I}_{\mathrm{L1L2L3}}\right]$$

- The equation is now very similar to that for symmetrical consumers
- Thus, the same transformations can be performed, so that here too there is a direct dependency between the original system and the transformation system

$$\left[\underline{Z}_{012}\right] = \begin{bmatrix} \underline{Z}_{\mathrm{S}} + 2 \cdot \underline{Z}_{\mathrm{K}} & 0 & 0 \\ 0 & \underline{Z}_{\mathrm{S}} - \underline{Z}_{\mathrm{K}} & 0 \\ 0 & 0 & \underline{Z}_{\mathrm{S}} - \underline{Z}_{\mathrm{K}} \end{bmatrix}$$

Symmetrical lines



- ESB of the three systems for symmetrical lines
- The dashed line is intended to reflect the neutral point treatment on the generation and consumption side for reference to the star voltages.